METRIC

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PERFORMANCE SPECIFICATION

CABLES, FIBER OPTICS, (METRIC), GENERAL SPECIFICATION FOR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the general requirements and characteristics for cables utilizing optical fibers for signal transmission.

1.2 Classification. Cables covered by this specification consist of one or more single fiber, optical transmission elements with properties as specified (see 3.1) and are of the following categories:

1.2.1 Fiber optic cable configuration type. The fiber optic cable configuration type is designated by a single number as indicated in table I.

TABLE I. Fiber optic cable configuration type.

Cable configuration type	Designation
Buffered fiber 1/	1
OFCC 2/	2
Cable bundle 3/	3
Ribbon 4/	4
Tube 5/	5

1/ The cable is constructed using buffered fibers (see 6.5.2) with no individual protective jackets or strength members.

2/ The cable is constructed using OFCCs (see 6.5.8) that are not grouped into bundles having bundle jackets or binders (see 6.5.4 and 6.5.1).

3/ The cable is constructed using buffered fibers or OFCCs that are grouped into cable bundles (see 6.5.3) with bundle jackets or binders.

4/ The cable is constructed using fiber ribbons (see 6.5.9).

5/ The cable is constructed using hollow tubes or ducts.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command (SEA 05G), DoD Standardization Program and Documents Division, Department of the Navy, 2531 Jefferson Davis Highway Arlington, VA 22206-5160, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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1.2.2 Class. The class designation is defined by the mode volume of the optical fiber.

MM - Multimode
SM - Single-mode

1.2.3 Application. The application designation is defined by the intended application of the optical cable.

A. Airborne
B. Shipboard
C. Ground benign
D. Space
E. Ground tactical

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in section 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

TT-P-645	-	Primer, Paint, Zinc-Molybdate, Alkyd Type.

DEPARTMENT OF DEFENSE

MIL-S-901	-	Shock tests, H.I. (High Impact), Shipboard Machinery, Equipment, and Systems, Requirements for.
MIL-PRF-49291	-	Fiber, Optical (Metric) General Specification for.

(See supplement 1 for list of associated specifications).

STANDARDS

FEDERAL

FED-STD-228	-	Cable and Wire, Insulated; Methods of Testing.

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitations.

NAVAL ENGINEERING STANDARDS

NES-713	-	Determination of the Toxicity Index of the Products of Combustion from Small Specimens of Material.

(Copies of Naval Engineering Standards are available from the preparing activity, Naval Sea Systems Command, Code 03J2, Arlington, VA 22206-5160.)

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NATIONAL AERONAUTICAL AND SPACE ADMINISTRATION (NASA)

NHB 8060.1	-	Flammability, Odor, and Offgassing and Compatibility Requirements and Test Procedures For Materials in Environments That Support Combustion.

(Application for copies should be addressed to Office of Safety and Mission Quality, (Code QR), NASA Headquarters, Washington, DC 20546.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM-E-84	-	Standard Test Method for Surface Burning Characteristics of Building Materials. (DoD adopted)
ASTM-D-470	-	Crosslinked Insulations and Jackets for Wire and Cable, Methods of Testing. Materials from Outgassing in a Vacuum Environment. (DoD adopted)
ASTM-D-512	-	Standard Test Method for Chloride Ion in Water. (DoD adopted)
ASTM-E-595	-	Standard Test Method for Total Mass Loss and Collected Volatile Condensable Materials from Outgassing in a Vacuum Environment. (DoD adopted)
ASTM-E-800	-	Standard Guide for Measurement of Gases Present or Generated During Fires. (DoD adopted)
ASTM-D-1141	-	Standard Specification for Substitute Ocean Water. (DoD adopted)
ASTM-D-1179	-	Standard Test Method for Fluoride Ion in Water. (DoD adopted)
ASTM-D-1246	-	Standard Test Method for Bromine Ion in Water. (DoD adopted)
ASTM-D-2565	-	Standard Practice for Operating Xenon Arc-Type (Water Cooled) Light-Exposure Apparatus With and Without Water for Exposure of Plastics. (DoD adopted)
ASTM-D-3761	-	Standard Test Method for Total Fluorine in Coal by the Oxygen Bomb Combustion/Ion Selective Electrode Method. (DoD adopted)
ASTM-D-3868	-	Standard Test Method for Fluoride Ions in Brackish Water, Seawater, and Brines.
ASTM-D-3869	-	Standard Test Method for Iodide and Bromide Ions in Brackish Water, Seawater, and Brines.
ASTM-D-4327	-	Standard Test Method for Anions in Water by Chemically Suppressed Ion Chromatography.

(Application for copies of ASTM Publications should be addressed to the American Society for Testing and Materials, 100 Bar Harbor Drive, Conshohocken, Pennsylvania, 19428-2959

ELECTRONIC INDUSTRIES ASSOCIATION/TELECOMMUNICATIONS INDUSTRY ASSOCIATION (EIA/TIA)

EIA/TIA-455	-	Standard Test Procedure for Fiber Optic Fibers, Cables, Transducers, Sensors, Connecting and Terminating Devices, and Other Fiber Optic Components. (DoD adopted)
EIA/TIA-455-3	-	Procedure to Measure Temperature Cycling Effects on Optical Fibers, Optical Cable, and Other Passive Fiber Optic Components. (DoD adopted)
TIA/EIA-455-4	-	Fiber Optic Component Temperature Life Test. (DoD adopted)
TIA/EIA-455-5	-	Humidity Test Procedure for Fiber Optic Components. (DoD adopted)
EIA/TIA-455-12	-	Fluid Immersion Test for Fiber Optic Components.
TIA/EIA-455-13	-	Visual and Mechanical Inspection of Fiber Optic Components, Devices, and Assemblies. (DoD adopted)

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TIA/EIA-455-20	-	Measurement of Change in Optical Transmittance. (DoD adopted)
TIA/EIA-455-25	-	Repeated Impact Testing of Fiber Optic Cables and Cable Assemblies. (DoD adopted)
TIA/EIA-455-32	-	Fiber Optic Circuit Discontinuities. (DoD adopted)
EIA-455-33	-	Fiber Optic Cable Tensile Loading and Bending Test. (DoD adopted)
EIA/TIA-455-37	-	Low or High Temperature Bend Test for Fiber Optic Cable. (DoD adopted)
EIA/TIA-455-39	-	Fiber Optic Cable Water Wicking Test. (DoD adopted)
TIA/EIA-455-41	-	Compressive Loading Resistance of Fiber Optic Cables. (DoD adopted)
EIA/TIA-455-42	-	Optical Crosstalk in Fiber Optic Components. (DoD adopted)
EIA/TIA-455-46	-	Spectral Attenuation Measurement for Long-Length, Graded-Index Optical Fibers. (DoD adopted)
EIA-455-50	-	Light Launch Conditions for Long-Length Graded-Index Optical Fiber Spectral Attenuation Measurements. (DoD adopted)
TIA/EIA-455-56	-	Test Method for Evaluating Fungus Resistance of Optical Fiber and Cable. (DoD adopted)
TIA/EIA-455-71	-	Procedure to Measure Temperature-Shock Effects on Fiber Optic Components.
EIA/TIA-455-78	-	Spectral-Attenuation Cut Back Measurement for Single-Mode Optical Fibers. (DoD adopted)
EIA/TIA-455-81	-	Compound Flow (Drip) Test for Filled Fiber Optic Cable. (DoD adopted)
EIA/TIA-455-84	-	Jacket Self-Adhesion (Blocking) Test for Fiber Optic Cable. (DoD adopted)
EIA/TIA-455-87	-	Fiber Optic Cable Knot Test. (DoD adopted)
EIA-455-88	-	Fiber Optic Cable Bend Test. (DoD adopted)
TIA/EIA-455-91	-	Fiber Optic Cable Twist-Bend Test. (DoD adopted)
TIA/EIA-455-98	-	Fiber Optic Cable External Freezing Test.
TIA/EIA-455-104	-	Fiber Optic Cable Cyclic Flexing Test. (DoD adopted)
EIA/TIA-455-190	-	Low Air Pressure (High Altitude) Testing of Fiber Optic Components.
TIA/EIA-598	-	Optical Fiber Cable Color Coding. (DoD adopted)

(Application for copies should be addressed to the Electronic Industries Association/Telecommunications Industry Association, 2500 Wilson Boulevard, Suite 300, Arlington, Virginia 22201-3834).

INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS

IEEE-STD-299	-	Standard for Measuring the Effectiveness of Electromagnetic Shielding Enclosures. (DoD adopted)

(Copies of IEEE documents are available from the IEEE, 445 Hoes Lane, PO Box 1331, Piscataway, NJ 08855-1331.)

INTERNATIONAL ELECTROTECHNICAL COMMISSION

IEC-794-1-E10	-	Optical fibre cables - Part 1: Generic specification.

(Copies of IEC documents are available from the U. S. National Committee of the IEC, 11 West 42nd Street, 13th Floor, New York, NY 10036.)

UNDERWRITERS LABORATORY (UL)

UL-910	-	Standard for Safety Test for Flame Propagation and Smoke Density Values for Electrical and Optical Fiber Cables Used in Spaces Transporting Environmental Air. (DoD adopted)
UL-1685	-	Standard for Vertical-Tray Fire-Propagation and Smoke-Release Test for Electrical and Optical-Fiber Cables.

(Application for copies should be addressed to: Underwriters Laboratories Inc., 333 Pfingsten Road, Northbrook, IL 60062).

(Non-Government standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated specifications or specification sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheet. In the event of any conflict between the requirements of this specification and the specification sheet, the latter shall govern.

3.2 Qualification. The optical fiber cables furnished under this specification shall be products that are authorized by the qualifying activity for listing on the applicable qualified products list before contract award (see 4.5 and 6.3). The provisions of 4.5.5 for retention of qualification are included in this requirement.

3.3 Materials. The cable shall be constructed of material as specified (see 3.1). Materials selected for cable usage shall be of a type and quality to assure compliance with the requirements of this specification, and shall be physically and chemically compatible for their intended use and throughout their intended lifetime. All combinations of materials used shall be nonnutrient to fungus and of a virgin nature. When specified (see 3.1), materials and combinations of materials used in cable construction shall meet the requirements for toxic or explosive fumes produced by flame. Materials and combinations of materials used in cable construction shall have no adverse effect on the health of personnel or equipment when used for its intended purpose.

3.3.1 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.3.2 Filler. The fillers shall be evenly distributed, easily removable, and shall provide compactness and cross-sectional circularity to the finished cable.

3.3.2.1 Waterblocking material. Waterblocking material used in cable interstices shall be compatible with all other cable materials. Unless otherwise specified, the material shall be clean, nontacky, and nonirritating to the touch when not exposed to moisture. The material shall be free-stripping from the cable and components by hand and shall not require the use of chemicals or other mechanical means of removal. The material shall not interfere with any termination technique used with finished cable or components.

3.3.3 Strength members. Unless otherwise specified, strength members shall be all dielectric (see 3.1).

3.3.4 Jackets. Materials used for jackets (inner and outer) shall be as specified (see 3.1).

3.3.5 Cable nuclear vulnerability. When specified (see 3.1), materials shall meet the nuclear vulnerability requirements specified in appendices to the individual specification sheets.

3.3.6 Materials for space applications. When specified, materials used in space flight applications shall meet the following additional material requirements.

3.3.6.1 Thermal vacuum outgassing. When specified and when tested in accordance with 4.8.5.1, all non-metallic materials shall not exhibit greater than 1.0 percent total mass loss and shall not produce greater than 0.1 percent collected volatile condensable materials.

3.3.6.2 Material flammability. When specified and when tested in accordance with 4.8.5.2, non-metallic materials shall be considered non-flammable or self-extinguishing if, less than 6 inches of the minimum use thickness sample is consumed and the time of burning does not exceed 10 minutes. There shall be no sparking, sputtering, or dripping of flaming particles from the test sample.

3.3.6.3 Material toxicity (offgassing). When specified and tested in accordance with 4.8.5.3, all non-metallic materials shall not exceed a total hazard index of 0.5.

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3.4 Design and construction. The design, construction, and physical dimensions of the complete cable and cable elements shall be as specified (see 3.1). The cable shall consist of one or more optical transmission elements individually surrounded by protective layers to provide performance and dimensional characteristics as specified (see 3.1). Unless otherwise specified (see 3.1), the cable shall be of a circular cross-section and concentric geometry. The surface of the jacket shall be dry and free from any coating, film, or treatment which would interfere with the bonding to the jacket of encapsulating or molding materials used in splicing and terminating. Unless otherwise specified (see 3.1), finished cable and optical fiber cable components (OFCC) short term (dynamic) minimum bend diameter (see 6.5.10) shall be as listed in table II. The cable and the cable components shall be designed to comply with the specified properties while subject to specific operating and storage temperature ranges as specified (see 3.1). The long term (static) minimum bend diameter (see 6.5.7) shall be as specified (see 3.1).

TABLE II. Minimum short term bend diameter.

Application	Bend diameter to cable diameter (or OFCC diameter)
Ground tactical (T) Shipboard (B) Ground benign (G)	5X 8X 24X

3.4.1 Optical fibers. Optical fibers shall be in accordance with MIL-PRF-49291 and the applicable specification sheets (see 3.1). The optical fiber shall be coated with a material to preserve the high, pristine tensile strength of the glass fiber. The fiber and coatings shall maintain their physical characteristics at temperatures not greater than 85°C. Coating and buffer material shall be readily removable by mechanical means.

3.4.1.1 Number of fibers. The number of optical fibers in the cable shall be as specified (see 3.1).

3.4.2 Cable configuration. The detailed cable configuration shall be as specified (see 3.1) and shall be either a buffered fiber, an OFCC, a cable bundle, a ribbon cable, or a tube cable (see 1.2.1).

3.4.2.1 Buffered fibers. Unless otherwise specified (see 3.1), the outer diameter of the buffered fiber shall be 900 μm. The buffer design shall be tight buffer, and the concentricity of the buffer shall be greater than or equal to 0.65 for 900 μm buffered fibers.

3.4.2.1.1 Buffered fiber color coding. Unless otherwise specified (see 3.1), individual fiber buffers shall be color-coded for identification in accordance with EIA/TIA-598. The fiber buffer within an OFCC shall be color coded the same as the OFCC jacket color code.

3.4.2.2 Optical fiber cable component (OFCC). The OFCC dimensions and concentricity requirements shall be as specified (see 3.1).

3.4.2.2.1 OFCC jacket color coding. Unless otherwise specified (see 3.1), individual OFCC jackets shall be color coded for identification in accordance with EIA/TIA-598. For cable designs with more than 12 OFCCs within a bundle, the OFCCs may be marked for identification in lieu of color coding.

3.4.2.2.2 OFCC kink. When tested in accordance with 4.7.3.5, visual inspection shall reveal no kinking, cracking, splitting, tearing, or collapsing of the OFCC jacket for free form loop diameters not less than the minimum short term bend diameter of the OFCC.

3.4.2.3 Cable bundle jacket. The cable bundle jacket dimensions and concentricity requirements shall be as specified in the specification sheet (see 3.1).

3.4.2.3.1 Cable bundle jacket color coding. Unless otherwise specified (see 3.1), individual cable bundle jackets shall be color coded for identification in accordance with EIA/TIA-598.

3.4.2.4 Optical fiber ribbon (Type 4 only). The dimensional requirements and the number of fibers per ribbon for the optical fiber ribbons shall be as specified (see 3.1). A ribbon shall consist of a linear array of nominally contiguous fibers which are held between the adhesive faces of two pressure sensitive tapes or are bonded together with a matrix material. The ribbons shall have no crossovers, defective fibers or splices.

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3.4.2.4.1 Ribbon fiber color coding. Unless otherwise specified (see 3.1), each fiber in a ribbon shall be color coded for identification in accordance with EIA/TIA-598.

3.4.2.4.2 Ribbon color coding. Ribbon color coding shall be as specified (see 3.1).

3.4.2.5 Binders. The dimensional requirements and the number of fibers, buffered fibers, or OFCC's for each binder shall be as specified (see 3.1).

3.4.2.5.1 Binder color coding. Unless otherwise specified (see 3.1), each binder shall be color coded for identification in accordance with EIA/TIA-598.

3.4.2.6 Cable tubes. The dimensional requirements for cable tubes shall be as specified (see 3.1).

3.4.2.6.1 Cable tube color coding. Cable tube color coding shall be as specified (see 3.1).

3.4.3 Cable jacket. The cable jacket shall provide environmental and physical protection to the enclosed cable elements. The cable jacket shall be applied concentrically to the cable core to maintain circularity in the completed cable. Jacket dimensions and dimensional tolerances shall be as specified (see 3.1). The concentricity of the jacket shall be equal to, or greater than 0.65. This requirement applies to primary layers, coatings, or underlying jackets. The jackets shall be easily removable without damage to other cable component members as specified in 3.6.18. The jackets shall be dry and free from any coating, film, or treatment that would tend to interfere with the bonding of it to encapsulating or molding materials used in splicing and terminating. All jackets shall be free of pinholes, blowouts, and bumps (see 4.7.2).

3.4.3.1 Cable and cable core component jacket colors. Unless otherwise specified (see 3.1 and 6.2), the color of the overall cable jacket shall be black. The individual OFCC jackets and cable bundle jackets shall be separately color-coded as specified in 3.4.2.2.1 and 3.4.2.3.1, respectively. The ribbon fibers and ribbons shall be color-coded as specified in 3.4.2.4.1 and 3.4.2.4.2, respectively.

3.4.4 Fillers. Fillers may be used to provide firmness, roundness, and water tightness of finished cables. Fillers shall be made of electrically nonconducting material meeting the applicable requirements of 3.5, 3.6 and 3.7. Filler material shall be of a consistency so as to not induce attenuation during the cabling process and shall have physical properties so as to prevent changes in optical parameters when the cable is subjected to the physical and environmental tests specified (see 3.1). The fillers shall be removable in accordance with 3.6.18.

3.4.5 Strength and central members. The strength members shall consist of peripheral layers of nonelectrically conducting materials as specified (see 3.1). If a central member is specified (see 3.1), the central member shall be electrically nonconductive.

3.4.6 Cabling. Optical cables shall be cabled as specified (see 3.1). The length of lay shall be that required to meet the minimum specified bend radius, flexing, and twist-bending requirements. Strength members, fillers, and central members may contain splices. Splices in the strength members, fillers, and central members shall be dimensionally indistinguishable within the manufacturer's tolerances from the unspliced components. The splice strength shall be not less than the strength of the unspliced material.

3.4.6.1 Cable physical dimensions. When tested in accordance with 4.7.3.1 and 4.7.3.2, the physical dimensions of the finished cable shall be as specified (see 3.1).

3.4.6.2 Concentricity. When tested in accordance with 4.7.3.3, the concentricity of the finished cable, OFCC, buffered fiber(s), and cable bundle(s), shall be as specified (see 3.1). The concentricity shall be greater than or equal to 0.65, and shall apply to jackets and underlying jackets.

3.4.6.3 Cable mass per unit length. When tested in accordance with 4.7.3.4, the mass per unit length of a fully assembled cable shall not exceed the value specified (see 3.1).

3.4.6.4 Cable continuous lengths. Lengths shall be as specified (see 6.2). Fully assembled cable shall be continuous and shall not be repaired or spliced.

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3.5 Optical performance requirements.

3.5.1 Attenuation rate. The attenuation rate of the fiber prior to cabling or application of other stresses shall be tested in accordance with 4.7.4.1. Unless otherwise specified (see 3.1), the change in attenuation rate of the optical fiber in the finished cable shall not exceed the values shown in Table III from the precabled coated fiber value. The attenuation rate at the wavelengths of operation of the finished cable shall be as specified (see 3.1).

Table III. Maximum change in attenuation rate.

Application	Fiber type	Change in attenuation rate (dB/km)
Tactical (T)	Multimode Single mode	$\leq$ 0.5 dB/km $\leq$ 0.3 dB/km
Shipboard (B)	Multimode Single mode	$\leq$ 1.0 dB/km $\leq$ 0.5 dB/km

3.5.2 Change in optical transmittance. The change in optical transmittance of the cabled fibers due to exposure to mechanical (environmental and physical) tests shall not exceed the values specified in table IV when tested in accordance with 4.7.4.2. The end resultant attenuation due to cumulative environmental and mechanical testing shall not exceed that specified (see 3.1).

TABLE IV. Maximum change in optical transmittance.

Fiber type	Change in optical transmittance	
	Mechanical (dB)	Environmental (dB/km) 1/
Multimode (MM) Single mode (SM)	0.5 0.2	0.5 0.3

1/ The change in optical transmittance requirement for freezing water immersion, gas flame, and shock are for the specified test length and shall not be normalized for one kilometer.

3.5.3 Crosstalk. When specified (see 3.1), the cable shall be tested in accordance with 4.7.4.3. The far end crosstalk between any two fibers shall be not greater than minus 60 dB below the active fiber optical output level.

3.6 Mechanical performance requirements.

3.6.1 Tensile loading and elongation. After testing a specimen of finished cable or OFCC in accordance with 4.7.5.1, there shall be no evidence of cracking, splitting, or breaking of the cable components or elongation greater than 2 percent. Unless otherwise specified, the change in optical transmittance of each fiber shall not exceed 2.0 dB during the test and the values specified in 3.5.2 after testing.

3.6.1.1 Operating tensile load. When specified (see 3.1), cable shall be tested in accordance with 4.7.5.1.1. There shall be no evidence of cracking, splitting, or breaking of the cable components. The change in optical transmittance shall not exceed the values specified in 3.5.2 during or after the test.

3.6.2 Dynamic bend. When tested in accordance with 4.7.5.2, the finished cable or OFCC shall reveal no jacket softening, surface damage (cracking, splitting, or other defect to permit jacket penetration), or identification marking impairment. The change in optical transmittance of each fiber shall not exceed the values specified in 3.5.2 during and after the test.

3.6.3 Low temperature flexibility (cold bend). When tested in accordance with 4.7.5.3, a post-test visual examination of the cable jacket shall reveal no cracking, splitting, or other failure permitting jacket penetration. The change in optical transmittance of each fiber shall not exceed the values specified in 3.5.2 after the test.

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3.6.4 Cyclic flexing. When tested in accordance with 4.7.5.4, a post-test visual examination of the cable shall reveal no splitting, cracking, or crazing of the specimen jacket. The change in optical transmittance of each fiber shall not exceed the values specified in 3.5.2 during and after the test.

3.6.5 Crush. When tested in accordance with 4.7.5.5, a post-test visual examination of the cable jacket shall reveal no cracking, splitting, or other defects to permit jacket penetration. Cable exterior deformation shall not be considered as damage or cable failure. The change in optical transmittance shall not exceed the values specified in 3.5.2 during and after the test and the optical crosstalk shall meet the requirements of 3.5.3 during and after the test.

3.6.6 Cable twist-bending. When tested in accordance with 4.7.5.6, finished cable specimens shall exhibit no evidence of degradation of jacket materials. The change in optical transmittance of each fiber shall not exceed the values specified in 3.5.2 during and after testing. A post-test visual examination shall reveal no jacket softening, surface damage (cracking, splitting, or other defect to permit jacket penetration), or any identification marking impairment which affects legibility.

3.6.7 Radial compression (for application B only). When specified and when tested in accordance with 4.7.5.7, the finished cable shall reveal no cracking, splitting, or other defects to permit jacket penetration. The change in optical transmittance of each fiber shall not exceed 0.1 dB during and after the test.

3.6.8 Impact. When tested in accordance with 4.7.5.8, the cable shall meet the following requirements: A visual examination of the cable jacket shall reveal no cracking, splitting, or other defect to permit jacket penetration. The jacket may crack or split on any low temperature impact after the second low temperature impact. The change in optical transmittance shall not exceed the values specified in 3.5.2 after the test.

3.6.9 Corner bend. Unless otherwise specified (see 3.1), cables shall be tested in accordance with 4.7.5.9. A post-test visual examination shall reveal no cracking, splitting, or tearing of the cable. The change in optical transmittance shall not exceed the values specified in 3.5.2 during and after the test.

3.6.10 Knot. When specified (see 3.1), the cable shall be tested in accordance with 4.7.5.10. A post-test visual inspection shall reveal no cracking, splitting, or tearing of the cable. The change in optical transmittance shall not exceed the values specified in 3.5.2 during and after the test.

3.6.11 Pressure cycling (for application B only). When specified (see 3.1), during the test specified in 4.7.5.11, the change in optical transmittance for each fiber shall not exceed the values specified in 3.5.2.

3.6.12 Hosing (for application B only). When specified and when tested in accordance with 4.7.5.12, the requirements of 3.6.12.1 and 3.6.12.2 apply only to water-blocked cables.

3.6.12.1 Low pressure. When tested in accordance with 4.7.5.12.1, water leakage through the cable specimen shall be only axial and shall not exceed 35 milliliter (ml).

3.6.12.2 Hydrostatic. When tested in accordance with 4.7.5.12.2, the cable specimen shall permit no more than 1000 ml times the squared cable outer diameter (in centimeters) flow of axial water and no more than 20.0 mm slippage of cable internal parts (strength members, OFCCs, fillers etc.). Splitting or cracking of the cable between the gland nuts (inside of the stuffing tube) is allowed.

3.6.13 Dripping. When specified (see 3.1), the cable shall be tested in accordance with 4.7.5.13. A post-test visual inspection of the lower vertical cable end shall reveal no evidence of globule formation nor shall any dripping be observed on the drip collector.

3.6.14 Cable jacket tear strength. When tested in accordance with 4.7.5.14, the cable jacket tear strength shall be not less than 60 newtons per centimeter (N/cm) of jacket thickness.

3.6.15 Cable jacket material tensile strength and elongation. When tested in accordance with 4.7.5.15, the tensile strength of the cable and cable component jacket materials shall be not less than 900 N/cm^2. The percent elongation shall be not less than 125 percent.

3.6.16 Cable abrasion resistance. When tested in accordance with 4.7.5.16, the cable shall withstand 250 cycles of scraping abrasion and 150 cycles of cable-to-cable abrasion. Unless otherwise specified, exposure of any layers below the cable jacket shall be considered failure.

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3.6.17 Cable shrinkage. When tested in accordance with 4.7.5.17, the total shrinkage of the length of finished cable (and buffered fiber, OFCC, cable bundle jacket and ribbon, as specified (see 3.1)) shall be not greater than 6.35 mm for thermoset jacket cables and 20 mm for thermoplastic jacket cables.

3.6.18 Cable element removability. When tested in accordance with 4.7.5.18, finished cable jacket, OFCC jacket, cable bundle jacket, optical fiber buffer, and ribbon tape shall be easily and cleanly removable by mechanical means without damage to the cable or optical fibers or both. No surface scratches or defects to the optical fiber shall be visible under 10X magnification after the fiber buffer material has been removed. The cable waterblock or filler materials, if applicable, shall be flexible and easily removable from any part to which it is in contact through the use of fingers only. The presence of occasional particles or slivers of filler residue will be acceptable, provided that these can be removed by light brushing with the fingers or with a dry cloth. Filler material which leaves residue that is removable only by vigorous wiping or through the use of solvents shall not be acceptable.

3.6.19 Durability of identification marking. Identification marking, including stripping or banding when specified (see 3.1), when applied to the outer surface of the finished cable, shall be capable of withstanding the durability test specified in 4.7.5.19 for 500 cycles. This test shall not be required when the identification marking is under a clear jacket.

3.6.20 Ribbon delamination (Type 4 only). When tested in accordance with 4.7.5.20, the ribbon shall not delaminate.

3.7 Environmental performance requirements. The finished cable shall perform according to all the requirements herein and as specified (see 3.1), during the specified operating environments and after the specified storage environment. The operating temperature range and storage temperature range shall be as shown in table V and as specified (see 3.1).

TABLE V. Temperature ranges.

Application	Operating range (^{O}C)	Storage range (^{O}C)
Ground tactical (T)	-46 to +71	-54 to +85
Shipboard (B)	-28 to +65	-40 to +70 1/
Ground benign (G)	-40 to +75	-40 to +75
Aircraft (A)	-46 to +125	-54 to +125
Space (S)	-60 to +85	-60 to +85

1/ The shipboard nonoperational temperature range is the same as the shipboard storage temperature range.

3.7.1 Temperature cycling. When tested in accordance with 4.7.6.1, a post-test visual examination shall reveal no jacket softening, surface damage, or identification marking impairment. Post-test cable outer diameter shall remain within plus or minus 10 percent of the pretest cable outer diameter. The change in optical transmittance of each fiber shall not exceed the values specified in 3.5.2 at the low and high temperatures during and after the test.

3.7.2 Thermal shock. When tested in accordance with 4.7.6.2, a post-test visual examination shall reveal no jacket softening, surface damage (cracking, splitting or other defects to permit jacket penetration), or identification marking impairment. Post-test cable outer diameter shall remain within 10 percent of the pretest cable outer diameter. The change in optical transmittance of each fiber shall not exceed the values specified in 3.5.2 after the test.

3.7.3 Temperature humidity cycling. When tested in accordance with 4.7.6.3, a post-test visual examination shall reveal no jacket softening, surface damage, or identification marking impairment. Post-test cable outer diameter shall remain within plus or minus 10 percent of the pretest cable outer diameter. The change in optical transmittance of each fiber shall not exceed the values specified in 3.5.2 during and after the test.

3.7.4 Storage temperature. When specified (see 3.1), the cable shall withstand the test specified in 4.7.6.4. The exterior surface of the test specimens shall show no cracks or defects. The change in optical transmittance of each fiber shall not exceed the values specified in 3.5.2 after the test.

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3.7.5 Barometric pressure (reduced). When specified (see 3.1), the cable shall withstand the test specified in 4.7.6.5. The change in optical transmittance of each fiber shall not exceed the values specified in 3.5.2 during and after the test.

3.7.6 Life aging. When tested in accordance with 4.7.6.6, a post-test visual examination shall reveal no jacket softening, surface damage, or identification marking impairment. Post-test cable outer diameter shall remain within plus or minus 10 percent of the pretest cable outer diameter. The jacket tensile strength and elongation shall be not less than 75 percent of the initial value. The change in optical transmittance of each fiber shall not exceed the values specified in 3.5.2 after the test.

3.7.7 Freezing water immersion (ice crush). When specified (see 3.1), the cable shall withstand the test specified in 4.7.6.7. The change in optical transmittance of each fiber shall not exceed the values specified in 3.5.2 during and after the test.

3.7.8 Weathering. When specified and tested in accordance with 4.7.6.8, after the 1200 hours exposure, the cable jacket shall show no evidence of softening, gumminess or surface damage (cracking, splitting, or other defects to permit jacket penetration). The jacket tensile strength and elongation shall be not less than 75 percent of the initial value (see 4.7.5.15).

3.7.9 Fluid immersion. When tested in accordance with 4.7.6.9, the tensile strength and elongation retention properties of the cable jacket material shall be not less than 50 percent of the initial values. The finished cable shall reveal no cracks, splits, gumminess, or voids in the cable jacket. The outer diameter of the finished cable shall not deviate greater than 50 percent. The cable jacket shall not be preconditioned before this test other than normal jacket processing.

3.7.10 Wicking. When specified for water tight cables (see 3.1) and when tested in accordance with 4.7.6.10, a post-test examination of the immersed end of the cable specimen shall reveal a water penetration of less than 5 millimeters (mm) from the cable end and a change in mass of less than 1 percent.

3.7.11 Jacket self-adhesion or blocking. When tested in accordance with 4.7.6.11, a post-test visual inspection shall reveal no areas of localized adhesion between contacting cable surfaces, the metal storage spool, or areas that are of a more severe nature than a "mild" condition.

3.7.12 Flammability. When tested in accordance with 4.7.6.12, the requirements of 3.7.12.1, 3.7.12.2, or 3.7.12.3 apply only when specified (see 3.1).

3.7.12.1 Flammability (60° angle). When tested in accordance with 4.7.6.12.1, cables shall self-extinguish within 30 seconds. The distance of flame travel shall not exceed 10 cm, and there shall be no ignition of the tissue paper.

3.7.12.2 Flame extinguishing. When tested in accordance with 4.7.6.12.2, cables shall be self-extinguishing and shall not burn to the top of the tray.

3.7.12.3 Smoke generation and flame propagation. When tested in accordance with 4.7.6.12.3, the cables shall meet the requirements of UL-910 for smoke generation and flame propagation.

3.7.13 Shock. When specified and tested in accordance with 4.7.6.13, the finished cable shall reveal no physical damage. Unless otherwise specified (see 3.1), discontinuities shall not have a magnitude greater than the values specified in 3.5.2 for change in optical transmittance and a duration greater than 50μs.

3.7.14 Water absorption. When tested in accordance with 4.7.6.14, the maximum water absorption of the finished cable or a sample of the external cable jacket material shall not be greater than 4.0 mg per square centimeter (mg/cm^2) of exposed cable surface area.

3.7.15 Paint susceptibility. When specified and tested in accordance with 4.7.6.15, the cable jacket shall show no signs of jacket weakening, cracking, or other damage.

3.7.16 Electromagnetic effects. When specified and tested in accordance with 4.7.6.16, the propagated radio frequency (RF) attenuation of the cable shall be not less than 100 dB attenuation for frequencies not greater than 1 GHz and 60 dB attenuation at 10 GHz.

3.8 Chemical properties.

3.8.1 Acid gas generation. When tested in accordance with 4.8.1, the acid gas generation, expressed as acid equivalent, shall be not greater than 2.0 percent of the weight of the sample.

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3.8.2 Halogen content. When tested in accordance with 4.8.2, the total halogen content of the cable shall be as specified (see 3.1).

3.8.3 Toxicity index. When tested in accordance with 4.8.3, the toxicity index shall not exceed 5.0.

3.8.4 Fungus resistance. When tested in accordance with 4.8.4, polymeric cable materials shall show sparse or very restricted microbial growth and reproduction with minor or inhibited substrate utilization. There shall be little or no chemical, physical, or structural change detectable.

3.9 Identification marking. Except as otherwise specified in the procurement contract or in the specification sheet, the finished cable shall be identified by a marking applied to the outer surface of the cable or visible through the outer surface. When cable is to be used in an end item for the Government, omission of the identification marking of product shall be permissible only when so stated in the specification sheet for the cable or the Government contract for the end item. The identification marking shall consist of the following, at intervals of 0.25 to 1.5 meters, as measured from the beginning of one complete marking to the beginning of the succeeding complete marking:

a. PIN (see 6.6).

b. Manufacturer's code designation.

c. The words "Fiber Optic Cable".

d. Date code (4 digit - week, year).

e. When specified (see 6.2), meter markings shall be included.

The identification marking shall be permanent and legible. Identification marking shall be applied with the vertical axis of the printed characters lengthwise of the cable when the nominal diameter of the finished cable is 1.25 mm or smaller. The vertical axis of the printed characters may be either crosswise or lengthwise of the cable when the nominal diameter of the cable exceeds 1.25 mm.

3.9.1 JAN and J marking. The United States Government has adopted and is exercising legitimate control over the certification marks "JAN" and "J", respectively, to indicate that items so marked or identified are manufactured to, and meet all the requirements of military specifications. Accordingly, items procured to, and meeting all of the criteria specified herein and in applicable specifications shall bear the certification mark "JAN" except that items too small to bear the certification mark "JAN" shall bear the letter "J". The "JAN" or "J" shall be placed immediately before the PIN except that if such location would place a hardship on the manufacturer in connection with such marking, the "JAN" or "J" may be located on the first line above or below the part number. Items furnished under contracts or orders which either permit or require deviation from the conditions or requirements specified herein or in applicable specifications shall not bear "JAN" or "J". In the event an item fails to meet the requirements of this specification and the applicable specification sheets or detail specifications, the manufacturer shall remove the "JAN" or the "J" from the sample tested and also from all items represented by the sample. The "JAN" or "J" certification mark shall not be used on products acquired to contractor drawings or specifications. The United States Government has obtained Certificate of Registration Number 504,860 for the certification mark "JAN" and Registration Number 1,586,261 for the certification mark "J".

3.10 Workmanship. All details of workmanship shall be in accordance with high grade fiber optic cable manufacturing practice. Cables shall be dimensionally uniform. The minimal level of visual examination to be performed shall be as specified in a through e and is not intended to restrict other pertinent workmanship examinations deemed necessary by the manufacturer (see 4.7.2).

a. Outer jacket shall be free of cuts, burnt areas, abrasions, holes, roughened areas, bulges, thin spots, and discontinuities.

b. Inner layers shall be free of cuts, holes, bulges, thin spots, and discontinuities.

c. Strength members shall be uniformly laid with no discontinuities.

d. Fillers and water sealant shall be uniformly distributed throughout the cable body.

e. OFCC's shall not kink at diameters above the minimum short term bend diameter.

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4. VERIFICATION

4.1 Verification program. Requirements for the verification program shall be as required by the qualifying activity (see 6.3.2).

4.2 Test equipment and inspection facilities. Requirements for test equipment and inspection facilities shall be as be as required by the qualifying activity (see 6.3.3).

4.3 Classification of inspections. The inspections specified herein are classified as follows:

a. Qualification inspection (see 4.5).

b. Conformance inspection (see 4.6).

4.4 Inspection conditions. Unless otherwise specified (see 3.1), all inspections shall be performed in accordance with the standard test conditions specified in EIA/TIA-455 or as specified herein. When a cable construction utilizes spliced fibers, test samples shall be selected to include the fiber splices.

4.5 Qualification inspection. Qualification inspection shall be performed at a laboratory acceptable to the Government, (see 3.2 and 6.3) on sample units produced with equipment and procedures normally used in production. Group qualification (extent of qualification) shall be as specified in 4.5.2.

4.5.1 Sample units. Unless otherwise specified (see 3.1), the sample units shall be as specified in table VI. The qualification sample shall consist of two sample units, one a minimum of 1 km in length and one a minimum of 0.5 km in length. The 1 km sample unit shall be cut to form two sample units, each a minimum of 0.5 km in length.

4.5.2 Extent of qualification. In instances where two or more specification sheets cover cables identical in materials and construction except for attenuation, or in instances where cable constructions include varying numbers of fibers, the cable configuration incorporating the lowest attenuation range and the largest number of fibers shall be submitted and shall meet all the requirements of this specification and the specification sheet (see 3.1). Qualification may be extended to include those cables with higher attenuation and fewer number of fibers provided the samples submitted in accordance with 4.5.1 meet all the requirements of the specification sheet.

4.5.2.1 Qualification by similarity for a modified cable jacket compound. Manufacturers who produce cables using one particular overall cable jacket compound and are qualified under the applicable specification sheets for cables and pass the size, attenuation rate, electromagnetic effects, low temperature flexibility (cold bend), cyclic flexing, impact, cable life, fungus, weathering, cross-link verification (if applicable), fluid immersion, paint susceptibility, jacket self adhesion or blocking, hosing, dripping, cable jacket tear strength, cable jacket tensile strength and elongation, cable abrasion resistance, cable shrinkage, durability of identification, smoke generation and flame propagation, flame extinguishing, water absorption, acid gas generation, halogen content, and toxicity index for cables with a modified overall cable jacket compound are qualified under the same specification sheets for cables with a modified overall cable jacket compound. Qualification by similarity testing shall be performed on the eight fiber cable for each modified overall cable jacket compound.

4.5.3 Inspection routine. The samples shall be subjected to the qualification inspection specified in table VI in the order shown. In cases where certain requirements and tests are applicable only when specified (see 3.1), these tests shall be conducted in the order shown when specified in the appropriate specification sheet (see 3.1). Tests which are specified as not applicable to a specific cable construction shall not be conducted. All sample units shall be subjected to the applicable inspections of groups I and II. Specimens shall be cut from each sample unit in lengths at least as long as specified in table VI. Test specimens from each sample unit shall be subjected to the applicable tests of group III through group V, inclusive, of table VI; however, each test specimen shall be subjected to only one group of tests in addition to groups I and II. Test specimens for group VI shall be cut from undamaged test specimens from groups III, IV, or V. Optical tests shall be performed on the sample when required in section 3 as specified by the individual test in section 4.

4.5.3.1 Qualification of modified designs. Qualification inspection of items which contain design changes from previously qualified items may be limited to a subset of the qualification inspections identified in 4.5.3. Qualification inspection reductions will be determined by the qualification activity based on the extensiveness of the design changes and the anticipated effects of those changes on the item performance.

4.5.4 Failures. One or more failures shall constitute qualification inspection failure.

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TABLE VI. Qualification inspection.

Inspection	Requirement paragraph	Test paragraph	Specimen length
Group I			
Visual and mechanical inspection	3.4, 3.9, 3.10	4.7.2	3 units 0.5 km each 3/
Attenuation rate	3.5.1	4.7.4.1	3 units 0.5 km each 4/
Group II			
Crosstalk	3.5.3	4.7.4.3	3 units 0.5 km each 4/
Electromagnetic effects	3.7.16	4.7.6.16	1 unit 6 m 5/
Group III			
Temperature cycling	3.7.1	4.7.6.1	2 units 0.5 km each 4/ (1 on reel, 1 off reel)
Temperature humidity cycling	3.7.3	4.7.6.3	2 units 0.5 km each 6/
Storage temperature	3.7.4	4.7.6.4	2 units 0.5 km each 6/
Low temperature flexibility (cold bend)	3.6.3	4.7.5.3	2 units 8 m each 7/
Cyclic flexing	3.6.4	4.7.5.4	6 units 5 m each 8/ (2 units for each temperature)
Crush	3.6.5	4.7.5.5	2 units 5 m each 7/
Cable twist-bending	3.6.6	4.7.5.6	6 units 5 m each 8/ (2 units for each temperature)
Impact	3.6.8	4.7.5.8	2 units 5 m each 7/
Barometric pressure (reduced)	3.7.5	4.7.6.5	1 unit .46 km 9/
Life aging	3.7.6	4.7.6.6	2 units 300 m each 7/
Tensile loading and elongation	3.6.1	4.7.5.1	2 units 150 m each 10/
Operating tensile loading	3.6.1.1	4.7.5.1.1	2 units 150 m each 11/
Freezing water immersion (ice crush)	3.7.7	4.7.6.7	2 units 30 m each 7/
Fungus resistance	3.8.4	4.8.4	2 units 0.5 m each 7/
Knot	3.6.10	4.7.5.10	6 units 5 m each 8/
Cable element removability	3.6.18	4.7.5.18	2 units 0.5 m each 7/
Flammability (60 degree angle)	3.7.12.1	4.7.6.12.1	2 units 6 m each 7/
Group IV			
Thermal shock	3.7.2	4.7.6.2	1 unit 0.49 km 4/ (on reel)
Weathering	3.7.8	4.7.6.8	1 unit 2 m 12/ and 3 material samples 13/
Fluid immersion	3.7.9	4.7.6.9	1 unit 2 m 12/ and 3 material samples 13/ for each specified fluid
Paint susceptibility	3.7.15	4.7.6.15	2 units 2 m 14/
Jacket self-adhesion or blocking	3.7.11	4.7.6.11	1 unit 30 m 12/
Shock	3.7.13	4.7.6.13	1 unit 30 m 12/
Dynamic bend	3.6.2	4.7.5.2	1 unit 150 m 15/
Hosing: Low pressure	3.6.12.1	4.7.5.12.1	1 unit 1.5 m 12/
Hydrostatic	3.6.12.2	4.7.5.12.2	1 unit 2 m 12/
Radial compression	3.6.7	4.7.5.7	1 unit 10 m 12/
Pressure cycling	3.6.11	4.7.5.11	1 unit 30 m 12/
Corner bend	3.6.9	4.7.5.9	2 units 5 m 14/
OFCC kink	3.4.2.2.2	4.7.3.5	2 units 1 m 12/

See footnotes at end of table.

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TABLE VI. Qualification inspection - Continued.

Inspection	Requirement paragraph	Test paragraph	Specimen length
Group V			
Dripping	3.6.13	4.7.5.13	1 unit 30 cm 12/
Cable jacket tear strength	3.6.14	4.7.5.14	3 units 1 m each 14/
Cable jacket material tensile strength and elongation	3.6.15	4.7.5.15	5 units 16/
Cable abrasion resistance	3.6.16	4.7.5.16	4 units 2 m each 17/
Cable shrinkage	3.6.17	4.7.5.17	3 units 0.5 m each 14/
Durability of identification	3.6.19	4.7.5.19	3 units 2 m each 14/
Ribbon delamination	3.6.20	4.7.5.20	3 units 2 m each 14/
Smoke generation and flame propagation	3.7.12.3	4.7.6.12.3	220 m 18/
Flame extinguishing	3.7.12.2	4.7.6.12.2	1 unit 50 m 12/
Wicking	3.7.10	4.7.6.10	2 units 2 m 14/
Water absorption	3.7.14	4.7.6.14	2 units 19/
Group VI			
Acid gas generation	3.8.1	4.8.1	1 unit 1 m 20/
Halogen content	3.8.2	4.8.2	1 unit 1 m 20/
Toxicity index	3.8.3	4.8.3	1 unit 1 m 20/
Thermal vacuum outgassing	3.3.6.1	4.8.5.1	1 unit 1 m 20/
Material flammability	3.3.6.2	4.8.5.2	1 unit 1 m 20/
Material toxicity (offgassing)	3.3.6.3	4.8.5.3	1 unit 1 m 20/

1/ Tolerance on 0.5 km length is plus or minus 10 percent provided results are normalized to 1 km.
2/ Tolerance on shorter lengths is plus or minus 10 percent.
3/ The visual and mechanical inspection shall only be conducted on a 2 m section of each sample.
4/ The same samples as used in the visual and mechanical inspection shall be used.
5/ A specimen cut from one of the visual and mechanical inspection test samples shall be used.
6/ The same samples as used in the temperature cycling test shall be used.
7/ A specimen cut from each sample used in the temperature cycling test shall be used.
8/ Three specimens cut from each sample used in the temperature cycling test shall be used.
9/ One of the samples used in the temperature cycling test shall be used.
10/ A specimen cut from each sample used in the life test shall be used.
11/ The same sample as used in the tensile loading and elongation test shall be used.
12/ A specimen cut from the sample used in the thermal shock test shall be used.
13/ Three jacket material samples as specified in 4.7.5.15 shall be used in this test.
14/ Specimens cut from the sample used in the thermal shock test shall be used.
15/ A specimen cut from the sample used in the thermal shock test or one of the samples used in the tensile loading and elongation test shall be used.
16/ Jacket material samples as specified in 4.7.5.15 shall be used in this test.
17/ Specimens cut from the sample used in the thermal shock test shall be used. Two specimens shall be used for scraping abrasion testing and two specimens shall be used in the cable to cable abrasion testing.
18/ A specimen consisting of several pieces, each cut from samples used in previous tests, shall be used.
19/ Jacket material samples as specified in 4.7.5.15 shall be used.
20/ A specimen cut from one of the samples used in the temperature cycling or thermal shock tests shall be used.
21/ Hybrid cables, consisting of alternating single mode and multimode fibers, can be used for qualification purposes. If a failure occurs for any fiber in a hybrid cable then all of the fibers in the cable are considered to have failed (both single mode and multimode).

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4.5.5 Retention of qualification. To retain qualification, the manufacturer shall verify in coordination with the qualifying activity the capability of manufacturing products which meet the performance requirements of this specification. Refer to the qualifying activity for the guidelines necessary to retain qualification to this particular specification. The manufacturer shall immediately notify the qualifying activity at any time that the inspection data indicates failure of the qualified product to meet the performance requirements of this specification.

4.6 Conformance inspection. Conformance inspection shall consist of the inspections and tests specified for group A inspection (table VII), group B inspection (table VIII), and group C inspection (table IX). Requirements for alternate forms of conformance inspection shall be as identified in the qualification instructions (see 6.3.4).

4.6.1 Inspection of product for delivery. Inspection of product for delivery shall consist of group A inspections.

4.6.1.1 Unit of product. A unit of product shall be 1 km of cable of the same part number. If a production run is less than a unit of product, then the quantity produced shall be one unit of product.

4.6.1.2 Production unit. The production unit shall consist of the number of units of product produced on the same production line or lines, and offered for inspection at one time. All of the units of product in the production unit submitted shall have been produced during the same production period with the same materials and processes.

4.6.1.3 Sample unit. A sample unit shall be a unit of product selected at random from the production unit without regard to quality.

4.6.1.4 Specimen. A specimen shall be an individual length of cable cut from the sample unit.

4.6.2 Group A inspection. Group A inspection shall consist of the inspections and tests specified in table VII. In cases where certain requirements and tests are applicable only when specified (see 3.1), these tests shall be conducted in the order shown when specified (see 3.1). Tests which are specified in the basic specification or specific slash sheet as not applicable to a specific cable construction shall not be conducted.

TABLE VII. Group A Inspection.

Inspection	Requirement paragraph	Test paragraph
Visual and mechanical	3.4, 3.9, 3.10	4.7.2
Attenuation rate	3.5.1	4.7.4.1

4.6.2.1 Sampling plan. Group A inspection shall be performed on 100 percent of delivered product.

4.6.2.2 Failures. One or more failures shall constitute group A inspection failure of the sample unit.

4.6.2.3 Disposition of sample units. Sample units that have failed any of the group A inspection tests shall not be shipped or submitted for group B testing.

4.6.3 Group B inspection. Group B inspection shall consist of the inspections specified in table VIII. In cases where certain requirements and tests are applicable only when specified (see 3.1), these tests shall be conducted in the order shown when specified (see 3.1). Tests which are specified in the basic specification or specific slash sheet as not applicable to a specific cable construction shall not be conducted. Group B inspection shall be made on sample units that have passed the group A inspection.

TABLE VIII. Group B Inspection.

Inspection	Requirement paragraph	Test paragraph
Low temperature flexibility (cold bend)	3.6.3	4.7.5.3
Dynamic bend	3.6.2	4.7.5.2

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4.6.3.1 Sampling plan. Two 170 m specimens shall be selected from those sample units covered by a single specification sheet within 24 months after the date of notification of qualification and during every 24 month period thereafter, except when the total production in a 24 month period is less than 2 units of product (2 km) or a total of 60 months have elapsed since the inspection was performed, in which case only one specimen shall be tested.

4.6.3.2 Failures. If one or more specimens fail to pass group B inspection, the production unit from which the specimens were selected shall be rejected.

4.6.3.3 Rejected production units. Requirements regarding the rework of rejected production units shall be as identified in the qualification instructions (see 6.3.10)

4.6.3.4 Disposition of sample units. Sample units from which a specimen has failed any of the group B inspection tests shall not be shipped, even though the production unit submitted is accepted.

4.6.4 Periodic inspection. Periodic inspection shall consist of group C. Except where the results of these inspections show noncompliance with the applicable requirements (see 4.6.4.1.4), delivery of products which have passed group A and group B inspections shall not be delayed pending the results of periodic inspection.

4.6.4.1 Group C inspection. Group C inspection shall consist of inspections specified in table IX. In cases where certain requirements and tests are applicable only when specified (see 3.1), these tests shall be conducted in the order shown when specified (see 3.1). Tests which are specified in the basic specification or specific slash sheet as not applicable to a specific cable construction shall not be conducted. Group C inspection shall be made on sample units selected from production units which have passed the group A and group B inspections.

4.6.4.1.1 Sampling plan. Two 500 m specimens shall be selected from those sample units covered by a single specification sheet within 60 months after the date of notification of qualification and during every 60 month period thereafter, except when the total production in a 60 month period is less than two units of product (2 km) inspection need not be made until production is at least 2 units of product.

4.6.4.1.2 Failures. One or more specimen or sample unit failures shall constitute group C inspection failure.

4.6.4.1.3 Disposition of sample units. Specimens that have been tested to group C inspection shall not be shipped.

4.6.4.1.4 Noncompliance. Requirements regarding failure of group C inspection shall be as identified in the qualification instructions (see 6.3.11).

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Table IX. Group C inspection.

Inspection	Requirement paragraph	Test paragraph
Group I		
Temperature cycling	3.7.1	4.7.6.1
Temperature humidity cycling	3.7.3	4.7.6.3
Cable twist-bending	3.6.6	4.7.5.6
Impact (low temperature)	3.6.8	4.7.5.8
Life aging	3.7.6	4.7.6.6
Tensile loading and elongation	3.6.1	4.7.5.1
Operating tensile load	3.6.1.1	4.7.5.1.1
Group II		
Knot	3.6.10	4.7.5.10
Cable element removability	3.6.18	4.7.5.18
Flammability (60 degree angle)	3.7.12.1	4.7.6.12.1
Fluid immersion	3.7.9	4.7.6.9
Hosing: Low pressure	3.6.12.1	4.7.5.12.1
Hydrostatic	3.6.12.2	4.7.5.12.2
Pressure cycling	3.6.11	4.7.5.11
Cable jacket tear strength	3.6.14	4.7.5.14
Cable abrasion resistance	3.6.16	4.7.5.16
Cable shrinkage	3.6.17	4.7.5.17
Ribbon delamination	3.6.20	4.7.5.20
Wicking	3.7.10	4.7.6.10
Group III		
Acid gas generation	3.8.1	4.8.1
Thermal vacuum outgassing	3.3.6.1	4.8.5.1
Material flammability	3.3.6.2	4.8.5.2
Material toxicity (offgassing)	3.3.6.3	4.8.5.3

4.7 Methods of inspection.

4.7.1 Equivalent test methods. The use of equivalent test methods is allowed provided that the preparing activity and the qualifying activity have approved the use of that equivalent test method by that manufacturer (see 6.3.4).

4.7.2 Visual and mechanical examination (see 3.4, 3.6, 3.7, 3.8, and 3.9 and 3.10 . Visual and mechanical examinations shall be performed in accordance with TIA/EIA-455-13 to verify that the design, construction, physical characteristics, dimensions, marking, and workmanship are in accordance with the applicable requirements. Visual examination shall be accomplished utilizing 10X magnification. Visual inspection for the color of the cable and the color of the fiber may be accomplished without magnification. Visual inspection of the cable may be limited to the cable ends and the exposed surface of the cable. Unspooling of the cable to inspect the unexposed portions is not required.

4.7.3 Fiber and cable construction inspections.

4.7.3.1 Cable and cable core component dimensions (see 3.4.6.1). Buffered fiber, OFCC, cable bundle, ribbon, and the finished cable dimensions shall be measured in accordance with FED-STD-228, method 1018.

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4.7.3.2 Finished cable diameter (see 3.4.6.1). The diameter of finished cable shall be computed from the circumference measurement determined in accordance with FED-STD-228, method 1441.

4.7.3.3 Concentricity (see 3.4.6.2). The concentricity of the buffer, OFCC jacket, cable bundle jacket, and finished cable jacket shall be determined by locating and recording the minimum and maximum wall thickness of the same cross section. The ratio of the minimum wall thickness to the maximum wall thickness is defined as concentricity.

4.7.3.4 Cable mass per unit length (see 3.4.6.3). The mass of 1 km of fiber optic cable shall be determined in accordance with FED-STD-228, method 8311.

4.7.3.5 OFCC kink (see 3.4.2.2.2). The OFCC shall be tested in a free form loop using method IEC 794-I-E10. Three specimens shall be tested from each fiber on each end of the cable. The free standing loop diameter shall be measured in a direction parallel to that in which the forces are applied. The test on each specimen shall be terminated when either a kink is formed or when one half the minimum bend diameter is reached. The free form loop diameter in which the first noticeable detrimental effect occurs is to be recorded along with the effect.

4.7.4 Optical properties inspections. Unless otherwise specified (see 3.1), the optical requirements specified herein shall be met at the center wavelength of 1.300 μm with a plus or minus 0.020 μm tolerance.

4.7.4.1 Attenuation rate (see 3.5.1). The multimode fiber attenuation rate shall be measured in accordance with EIA/TIA-455-46. For multimode fibers, the source shall be noncoherent. Light launch conditions used during the attenuation rate measurements shall be made in accordance with EIA-455-50 and table X. The attenuation rate of each individual single mode fiber shall be measured in accordance with EIA/TIA-455-78.

TABLE X. Light Launch conditions for attenuation rate tests.

Fiber type	Launch conditions
SM	30-mm diameter mandrel
MM	70/70 restricted or equivalent

4.7.4.2 Change in optical transmittance (see 3.5.2). This test shall evaluate the change in optical transmittance of the fibers due to exposure of the cable to one or more inspection (environment and physical) tests.

4.7.4.2.1 Method. The change in optical transmittance of each fiber shall be measured in accordance with TIA/EIA-455-20, utilizing a monitor fiber to evaluate the change in transmittance due to exposure of the cable to environmental and physical tests. Any optical power detection method may be utilized if the method has the sensitivity to measure the differential optical power levels as specified in the individual mechanical test requirements of section 3, and if the method provides repeatable readings (less than 3 percent variation). A pretest optical power measurement shall be made and the specimen shall then undergo inspection testing. All optical power measurements, subsequent to the pretest measurement, shall be referenced to the pretest value and the change in dB calculated.

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4.7.4.2.2 Guidelines. These types of measurements require highly stable optical devices (source and detector) and repeatability of loss at the device-to-fiber interface. Use of the same reference fiber for calibrating the light source power output just prior to making all the measurements on the cable specimen, will enhance the measurement accuracy.

4.7.4.3 Crosstalk (see 3.5.3). Fully assembled cable specimens shall be tested for crosstalk in accordance with EIA/TIA-455-42. When applicable, the crosstalk shall be determined between a minimum of three randomly selected fibers in a multi-fiber cable, using each selected fiber sequentially as the actively transmitting element, and measuring the far-end power output of all other fibers in the cable specimen. The center wavelength shall be 1.300 μm $\pm$ 0.020 μm, and the light launch conditions shall be as specified (see 4.7.4.1).

4.7.5 Mechanical properties inspections (see 3.6). Unless otherwise specified (see 3.1), for mechanical tests, the specimen length shall be as given in individual inspections. The change in optical transmittance shall be measured in accordance with 4.7.4.2. The change in optical transmittance shall be measured for all fibers in the cable or a total of 12 fibers, whichever is less. In large count fiber cables, at least one fiber per ribbon, cable bundle, or binder shall be monitored. Visual examination, where required, shall be conducted in accordance with 4.7.2.

4.7.5.1 Tensile loading and elongation (see 3.6.1). The tensile loading on both the OFCC and the finished cable shall be measured in accordance with EIA-455-33. The OFCC may be tested using the same mandrel size as is used to test the finished cable. A minimum preload of 45 newtons shall be placed on the cable. The load shall be increased to 2500 newtons times the cable outer diameter in centimeters in 4 equal increments. Attenuation measurements on all fibers shall be determined in unstressed and stressed conditions. At the completion of this test, the cable jacket shall be visually examined in accordance with 4.7.2.

4.7.5.1.1 Operating tensile load (see 3.6.1.1). The operating tensile load of the finished cable shall be tested in accordance with EIA-455-33. A minimum preload of 45 newtons shall be placed on the cable. The load shall be increased to 500 newtons times the cable outer diameter in centimeters and held for 72 hours. Attenuation measurements on all fibers shall be determined in unstressed and stressed conditions. At the completion of this test, the cable jacket shall be visually examined in accordance with 4.7.2.

4.7.5.2 Dynamic bend (see 3.6.2). The finished cable and OFCC shall be pulled 90° over a sheave whose outside diameter is equal to the minimum bend diameter of the cable, rounded to the next higher centimeter. A 90° bend shall be placed in the cable by the sheave. The cable shall be pulled at a rate of 9 meters per minute (m/min), with a minimum tensile load of 875 newtons for each centimeter of cable outer diameter. The cable core components shall be pulled and loaded as specified (see 3.1). Specimen length shall be 150 meters. Apparatus shall be used to permit pulling the entire specimen length over the sheave. Tensile load shall not be applied by friction devices in direct contact with the cable. Friction applied to the supply reel or spool is an acceptable technique. The change in optical transmittance shall be measured during and after the test. At the completion of the test, the cable jacket shall be visually examined in accordance with 4.7.2.

4.7.5.3 Low temperature flexibility (cold bend) (see 3.6.3). The low temperature flexibility characteristic shall be tested in accordance with procedure II of EIA/TIA-455-37. The following special test conditions shall apply:

a. The mandrel diameter shall be equal to the minimum bend diameter of the cable rounded up to the nearest centimeter.

b. The conditioning temperature shall be the low operating temperature plus or minus 2°C.

c. Tensioning masses as specified in EIA/TIA-455-37 shall be used.

d. Test level 6 shall apply (3 mandrel turns).

e. Visually examine the specimen under ten-power magnification.

f. Optical loss measurements at the low conditioning temperature prior to and after bending are not required. Post test loss or transmittance measurements are required after the sample has been returned to room temperature and the bends removed. The post test loss or transmittance values shall be compared to the loss or transmittance values obtained prior to the low temperature conditioning.

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4.7.5.4 Cyclic flexing (see 3.6.4). A length of finished cable shall be tested at 30 cycles per minute over a sheave whose outer diameter is equal to the cable minimum bend diameter rounded up to the nearest centimeter. The cyclic flexing test shall be performed in accordance with TIA/EIA-455-104. The test sample shall be conditioned at the test temperature before conducting each test for a duration not less than two hours. The change in optical transmittance shall be measured during and after the test. After completion of the test, the specimen shall be visually examined in accordance with 4.7.2. The test shall be conducted at the temperatures and for the number of cycles shown in table XI for the application specified.

TABLE XI. Cyclic flexing test limits.

Application	Number of cycles	Temperature
Ground tactical (T)	2000 2000 2000	Minimum operating temperature 25 ± 2°C Maximum operating temperature
Shipboard (B)	500 100	25 ± 2°C Minimum operating temperature

4.7.5.5 Crush (see 3.6.5). Compressive loading testing shall be accomplished in accordance with TIA/EIA-455-41 for the finished cable. Unless otherwise specified (see 3.1), the following special test conditions shall apply:

a. The compressive load exposure shall be not less than 2000 N/cm of cable outer diameter (in cm), held for 3 minutes, and released.

b. The compressive loading rate shall be not less than 2000 N/min.

c. The change in optical transmittance shall be measured while the specimen is under load and after load removal. Visual inspection of the specimen shall be made under 10X magnification after load removal.

d. A break in any fiber caused by this test shall be a failure of the cable.

e. Optical crosstalk shall be monitored in accordance with 4.7.4.3.

4.7.5.6 Cable twist-bending (see 3.6.6). A length of cable shall be tested over a sheave whose outside diameter is equal to the minimum bend diameter of the cable, rounded to the next higher centimeter for the number of cycles and at the temperatures listed in table XII for the specified application. Twist bending shall be accomplished in accordance with TIA/EIA-455-91. Unless otherwise specified (see 3.1), the test load shall be 100 newtons. The test sample shall be conditioned at the test temperatures before conducting each test for a duration not less than two hours. The change in optical transmittance shall be measured during and after the test. After the tests, the specimen shall be visually examined in accordance with 4.7.2.

Table XII. Cable twist bending test limits.

Application	Number of cycles	Temperature
Ground tactical (T)	2000 2000 2000	Minimum operating temperature 25 ± 2°C Maximum operating temperature
Shipboard (B)	500 100	25 ± 2°C Minimum operating temperature

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4.7.5.7 Radial compression (see 3.6.7). The intent of this test is to determine the response of the finished cable to the radial compressive forces that are applied to the cable when it is installed in multi cable penetrators (MCP). A cable specimen shall be fitted into three multi cable penetrators of appropriate size . A radial force shall be applied to the cable within the multi cable penetrators by installing the cable in accordance with the approved assembly drawing for the appropriate multi cable penetrator. The insert block inner diameter shall be not greater than 1 mm larger than the cable outside diameter. Multi cable penetrators that are an integral part of an interconnection box (see MIL-I-24728) may be used for this test. The minimum torque applied to the nuts on the MCP wedgepack shall be 5.6 Nm. Cable exterior deformation shall not be considered as danger or cable failure. The change in optical transmittance shall be measured during and after the test.

4.7.5.8 Impact (see 3.6.8). A length of cable specimen shall be tested in accordance with TIA/EIA-455-25. The specimen shall be conditioned at the test temperature for a duration not less than two hours before conducting each test. The change in optical transmittance shall be monitored after testing. During the low temperature test, after the second cycle, the test shall be temporarily halted and the cable jacket shall be visually examined in accordance with 4.7.2. At the completion of each test, the cable jacket shall be visually examined in accordance with 4.7.2. The test shall be conducted at the conditions listed in table XIII for the application specified.

TABLE XIII. Application test conditions.

Application	Number of cycles	Temperature
Ground tactical (T)	50 100 50	Minimum operating temperature 25 ± 2°C Maximum operating temperature
Shipboard (B)	50 20	25 ± 2°C Minimum nonoperating temperature

4.7.5.9 Corner bend (see 3.6.9). A length of finished cable shall be tested in accordance with EIA-455-88 except that the radius of the corner shall be 5 times the cable outer radius, rounded to the next highest centimeter (for cable only) and 1.3 cm for cable core components. A test force of 1000 N/cm of cable outer diameter or 500 N, whichever is less, shall be applied for one minute. The change in optical transmittance shall be measured after one minute of loading (while loaded) and after unloading.

4.7.5.10 Knot (see 3.6.10). A length of finished cable shall be tested in accordance with EIA/TIA-455-87, method A, except the test force shall be 1000 N/cm of cable outer diameter or 500 N, whichever is less.

4.7.5.11 Pressure cycling (see 3.6.11). The specimen shall be installed in a water pressure chamber with both ends brought out approximately 1 m through bulkhead stuffing tubes (MIL-S-24235 stuffing tubes are recommended). The protruding ends of the specimen shall be prepared for attenuation testing. The specimen within the chamber shall be subjected to 24 pressure cycles. During each cycle, the gauge pressure of the chamber shall be raised to 7.0 MPa (1,000 lbf/in^2) and reduced to zero. For each cycle, the high pressure shall be held for 30 minutes and the zero pressure shall be held for 30 minutes. The change in optical transmittance shall be measured continuously throughout the test on a minimum of three randomly selected fibers per specimen, when applicable.

4.7.5.12 Hosing (see 3.6.12). Hosing shall be as specified in 4.7.5.12.1 and 4.7.5.12.2.

4.7.5.12.1 Low pressure (see 3.6.12.1). A fully assembled cable specimen shall be tested in accordance with the procedure specified herein. One end of the specimen shall be placed in a terminal fitting which will allow water pressure to be applied directly to the exposed cross-sectional area of the end of the cable. Exposure of the sides of the cable to the water shall be kept to a minimum, and the fitting shall not exert radial compression against the cable. Unless otherwise approved by the qualifying activity, the sealer used for the packing and in the terminal fitting shall be a metal alloy having a maximum melting point of 88°C. The specimen shall be subjected to a sea water pressure of 172 kilopascal (kPa) (25 pounds per square inch (lbf/in^2)) for a period of 6 hours at 23°C. A 3 percent solution of salt (NaCl) water or sea water in accordance with ASTM-D-1141 shall be used. Water leakage through the specimen shall be measured and shall not exceed the maximum leakage specified in 3.6.12.1.

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4.7.5.12.2 Hydrostatic (see 3.6.12.2). A fully assembled cable specimen shall be tested in accordance with the procedure specified herein. One end of the specimen shall be fitted into the appropriate size of stuffing tube (refer to MIL-STD-2042 for guidance in choosing the appropriate size stuffing tube). The specimen end shall protrude from the stuffing tube a minimum of 0.5 m and a maximum of 0.75 m. The stuffing tube shall be torqued to 41 Nm and preconditioned at ambient temperature for 24 hours. The stuffing tube shall then be retorqued to 41 Nm and the cable internal components trimmed flush with the cable jacket end. Water pressure shall be applied to the stuffing tube end of the specimen. A 3 percent solution of salt (NaCl) water or sea water in accordance with ASTM-D-1141 shall be used. The water pressure shall be gradually applied up to the specified value (see 3.1) over a 3 to 10 minute time period and then held for 6 hours. Water leakage through the specimen and slippage of the cable internal parts shall not exceed the values specified in 3.6.12.2.

4.7.5.13 Dripping (see 3.6.13). A fully assembled cable specimen shall be tested in accordance with EIA/TIA-455-81. The specimen shall be suspended vertically in an oven for a period of 6 hours at an oven temperature of 150 $\pm$ 3°C. The extent of fluid flow or dripping from the open cable end shall be reported. Visual inspection of the specimen shall be made using ten-power magnification.

4.7.5.14 Cable jacket tear strength (see 3.6.14). The tear strength of the finished cable shall be determined in accordance with FED-STD-228, method 3111.

4.7.5.15 Cable jacket material tensile strength and elongation (see 3.6.15). Unless otherwise specified (see 3.1), extruded flat specimens of the jacket material shall be tested in accordance with FED-STD-228, method 3021 and 3031, with 2.5 cm bend marks, 2.5 cm jaw separation, and a rate of travel of 25 cm/minute. The thickness of the specimen shall be measured using a micrometer.

4.7.5.16 Cable abrasion resistance (see 3.6.16).

4.7.5.16.1 Cable scraping resistance.

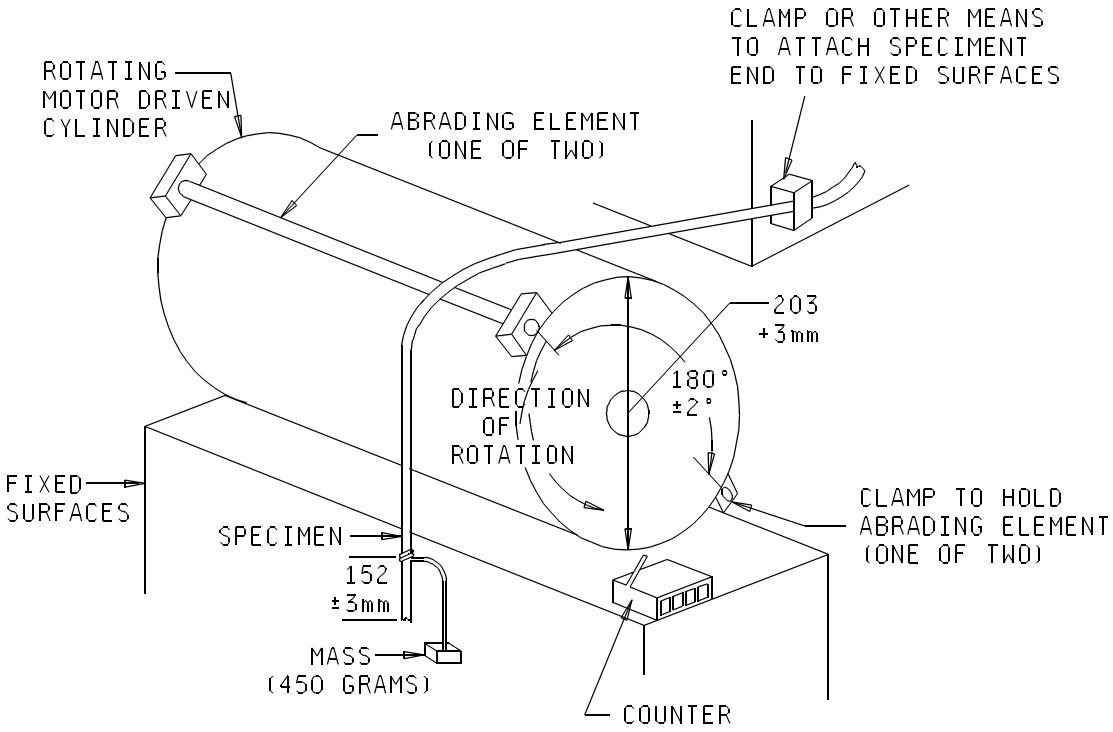
4.7.5.16.1.1 Test apparatus. The tester shall hold the test specimen firmly clamped in a horizontal position with the outer longitudinal surface of the specimen fully exposed. The tester shall rub an edge (a drum with an abrading edge, or equivalent, as shown in figure 1 shall be used) repeatedly over the outer surface of the specimen in such a position that the longitudinal axis of the edge and the specimen are at right angles to each other with the edge and outer surface of the specimen in contact. The clamped end of the cable shall be positioned such that the abrading element abrades the cable for the length specified. The abrading elements shall consist of high speed tool bits which have been ground on two adjacent longitudinal sides to produce a single, sharp 90 degree longitudinal edge, free of visible nicks. A weight affixed to the cable sample as shown in figure 1 shall control the force exerted normal to the surface of the jacket material. A motor-driven, reciprocating cam mechanism shall be used to deliver an accurate number of abrading strokes in a direction parallel to the longitudinal axis of the specimen. The number of cycles shall be measured by a counter. The length of the stroke shall be 5 cm and the frequency of the stroke shall be 30 cycles (60 strokes) per minute.

4.7.5.16.1.2 Test procedure. The specimen shall be clamped in the tester and a mass of 0.45 kg shall be carefully applied by the edge to the surface of the jacket. Two tests shall be performed on each specimen being moved forward 20 cm and rotated clockwise 90 degrees along the longitudinal axis between each test. Each test shall be discontinued when the specified number of cycles is attained for each of the two tests performed on each specimen.

4.7.5.16.2 Cable-to-cable abrasion.

4.7.5.16.2.1 Test apparatus. Cable-to-cable abrasion test apparatus and specimen relationship are shown on figures 2 and 3 respectively. The test apparatus shall consist of an electromagnetic transducer (driver) rigidly mounted on a heavy steel frame with the axial motion of the driver in a horizontal plane. The transducer shall drive a rocker arm via a spring steel bar. Mounted on this arm shall be a curved specimen holder upon which is mounted one of the cable (lower). The curvature of the upper surface of the specimen holder shall be an arc whose center is located at the pivot point of the rocker arm. The second (upper) specimen shall be mounted on the underside of a beam which is fastened to the frame through a thin, flexible nickel titanium alloy strip which shall serve as a hinge and allow the beam to be displaced only in a vertical direction. The beam and the driven specimen holder shall be positioned such that two cable specimens form an included angle of 60 degrees. The lower specimen holder shall be driven, by the transducer, at a rate of 1 Hertz and a peak-to-peak amplitude of 0.4 cm, along the bisector of the included angle. This symmetrical driving arrangement shall produce wear patterns of equal area on both specimens.

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Notes:

1. The length from the top center of the rotating cylinder to the end of the cable specimen to which the weight is attached shall be a minimum of 762 mm (30 inches).

2. The bottom edge of the clamped specimen shall be level with the top surface of the rotating cylinder.

FIGURE 1. Scraping abrasion test apparatus.

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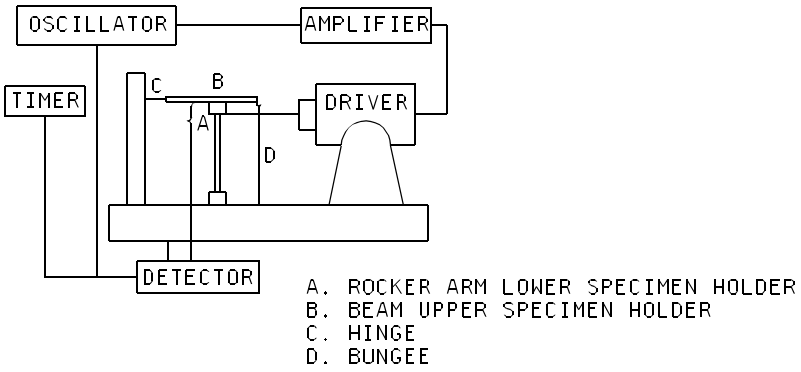


FIGURE 2. Abrasion resistance test apparatus.

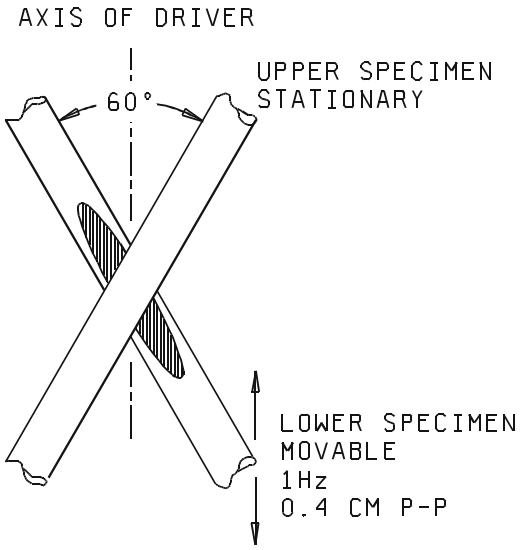


FIGURE 3. Abrasion resistance specimen relationships.

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4.7.5.16.2.2 Test procedure. Test specimens shall be mounted as described above. The applied force between the two cable specimens shall be produced by one or more rubber bungees between the beam and the frame. The actual force shall be measured with a force gauge directly in line with the intersection of the two specimens. The force measurement shall be taken when a force just sufficient to separate the two specimens is achieved. The force shall be measured before and after the test, and the results shall be averaged. The average force determined in this manner shall be 10.0 ± 1.0 newtons. The end point of the test shall be when the specified number of cycles is attained for the specimen under test.

4.7.5.17 Cable shrinkage (3.6.17). A 30 cm specimen of cable shall be cut so that all components are flush at both ends. The specimen shall be aged at 150°C ± 3°C for 6 hours in an air-circulating oven. At the end of this period, the specimen shall be removed from the oven and allowed to return to room temperature. The length of the specimen shall again be measured and the shrinkage shall be calculated as the difference between the two measurements.

4.7.5.18 Cable element removability (see 3.6.18). This test is applicable to the materials used to fill the voids within or between cable bundles, between the cable core components, or between and around fiber ribbons. It is also applicable to filler material used to fill the voids between the cable core components inner and outer jackets.

4.7.5.18.1 Specimens. A specimen shall consist of a length of completed cable approximately 1.0 meter long.

4.7.5.18.2 Procedure. All cable parts external to the cable core, such as cable jacket and strength member, shall be removed from one end of the specimen so as to expose the cable fibers and filler material for a distance of approximately 0.9 meter. Using fingers only (no hand tool), the filler shall be separated from the buffered fibers, cable core components, cable bundles, or fiber ribbon(s) for their fully exposed length. For cable bundles, approximately 40 cm of jacket shall be removed exposing the buffered fibers and cable core components, or both. The residual filler material on the buffered fiber or cable core component outer jackets, or both, shall be removed using fingers only. For core component, approximately 20 cm of the outer jacket shall be removed exposing the core component's core. The core component strength member shall be pushed back and removed exposing the inner jacket. The residual filler material on the core component inner jacket shall be removed using fingers only. For ribbons, approximately 20 cm of the tapes or matrix materials shall be removed from the ribbons and fibers shall be separated. Any residual material left on the fibers from the tapes shall be removed with a dry cotton cloth.

4.7.5.19 Durability of identification marking (see 3.6.19). The durability of product identification or markings applied to the cable for coding shall be evaluated at 20°C to 25°C.

4.7.5.19.1 Durability testing apparatus. The marking durability tester shall be designed to hold a short specimen of finished cable firmly clamped in a horizontal position with the upper longitudinal surface of the specimen fully exposed. This instrument shall be capable of rubbing a small cylindrical steel mandrel, 0.60 mm ± 0.05 mm in diameter, repeatedly over the upper surface of the cable, in such position that the longitudinal axes of the mandrel and the specimen are at right angles to each other with their cylindrical surfaces in contact. A mass affixed to a jig above the mandrel shall control the force exerted normal to the outer surface of the jacket. A motor-driven, reciprocating cam mechanism and counter shall be used to deliver an accurate number of abrading strokes in a direction parallel to the axis of the specimen. The length of the stroke shall be 10 mm and the frequency shall be 120 strokes (60 stroking cycles) per minute.

4.7.5.19.2 Durability testing procedure. In performing the test, a specimen of cable shall be mounted in the specimen clamp and the mass of 150 grams shall be applied through the abrading mandrel to the marked surface. The counter is then set at zero and the drive motor started. The test shall continue for 500 cycles. The specimen shall be removed and visually examined in accordance with 4.7.2. Failure is defined as an erasure or obliteration to the point of illegibility of any marking contacted during the test exposure.

4.7.5.20 Ribbon delamination (see 3.6.20). A 30 cm sample of fiber ribbon shall be exposed to a temperature of 88°C ± 2°C and relative humidity of 95 percent for 72 hours in a forced air oven.

4.7.6 Environmental tests (see 3.7). For environmental tests, the specimen lengths are given in the individual inspections. Change in optical transmittance measurements shall be made in accordance with 4.7.4.2. Measurements of the change in optical transmittance for environmental properties shall be made for all fibers in the cable or a total of 12 fibers, whichever is less. In large count fiber cables, at least two fibers per ribbon, cable bundle, or binder shall be tested. Visual examination of the specimen after exposure, where required, shall be conducted in accordance with 4.7.2.

4.7.6.1 Temperature cycling (see 3.7.1). The cable shall be tested in accordance with EIA/TIA-455-3 using the test condition schedule and soak times in accordance with table XIV. Change in optical transmittance shall be measured during and after the test.

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TABLE XIV. Temperature cycling steps.

Step	Temperature (^{O}C)	Duration (hours)
1. Maintain	Room ambient	8 (min)
2. Ramp to	Low operating temp +0, -3	2
3. Maintain	Low operating temp +0, -3	8 (min)
4. Ramp to	25 ± 2	2
5. Maintain	25 ± 2	6 (min)
6. Ramp to	High operating temp +3, -0	1
7. Maintain	High operating temp +3, -0	6 (min)
8. Ramp to	25 ± 2	1
9. Maintain	25 ± 2	6 (min)
10. Repeat steps 2 through 9 four additional times for a total of five cycles		

4.7.6.2 Thermal shock (see 3.7.2). Cables shall be tested in accordance with TIA/EIA-455-71 using test condition A-0. The temperature extremes shall be the specified storage temperature extremes (see 3.1). Soak times less than those specified in TIA/EIA-455-71 may be used if approved by the qualifying activity. The change in optical transmittance shall be measured after the test. Pretest and post-test measurements of the cable outer diameter shall be made and reported. Visual inspection of the test specimens shall be accomplished using ten-power magnification.

4.7.6.3 Temperature humidity cycling (see 3.7.3). Cables shall be tested in accordance with TIA-EIA-455-5 method B. The subcycle shall be included in the test. The change in optical transmittance shall be measured during and after the test. Pretest and post-test measurements of the cable outer diameter shall be made and reported. Visual inspection of the test specimens shall be accomplished using ten-power magnification.

4.7.6.4 Storage temperature (see 3.7.4). The specimen shall be subjected to 240 hours minimum of exposure to the low storage temperature extreme plus or minus 2^{O}C. Following the low temperature exposure, the specimen shall be exposed to room ambient temperature for a period of 24 hours, plus or minus 1 hour. The specimen shall then be exposed for 240 hours minimum to the high storage temperature extreme plus or minus 2^{O}C. Following the high temperature test, the specimen shall be exposed to room ambient temperature for a period of 24 hours, plus or minus 1 hour. The change in optical transmittance shall be measured after the test. After the test, the specimen shall be visually examined using the ten-power magnification.

4.7.6.5 Barometric pressure (reduced) (see 3.7.5). The cable shall be subjected to the test of EIA/TIA-455-190, test condition K, with mounting the same as that in normal service. Altitude exposure shall be 1 hour, minimum. The change in optical transmittance shall be measured following the test.

4.7.6.6 Life aging (see 3.7.6). Flat extruded jacket material shall be tested in accordance with TIA/EIA-455-4 at the temperature and duration listed in table XV for that particular material. Fully assembled cable samples shall be tested in accordance with TIA/EIA-455-4 for 240 hours at 110°C or for 450 hours at 100°C. The change in optical transmittance shall be measured after the test. The special test conditions specified in a through c shall apply.

a. Pretest and post-test measurements of the cable outer diameter shall be made and reported.

b. Visual inspection of the jacket shall be made using ten-power magnification.

c. The tensile strength and elongation of the aged extruded flat jacket material shall be determined in accordance with 4.7.5.15 after the test.

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TABLE XV. Cable life test conditions.

Jacket material	Aging conditions
Thermoplastic	240 hours @ 110°C
Thermoset	4 hours @ 175°C

4.7.6.7 Freezing water immersion (ice crush) (see 3.7.7). The specimen shall be tested in accordance with TIA/EIA-455-98, method A, procedure 1. The change in optical transmittance shall be monitored during and after the test.

4.7.6.8 Weathering (see 3.7.8). Flat extruded jacket material and the finished cable shall be tested in accordance with ASTM-D-2565, using conditions specified in table XVI. After completion of the test exposure, the finished cable shall be visually examined and the flat extruded jacket material shall be tested for jacket material tensile strength and elongation in accordance with 4.7.5.15.

TABLE XVI. Conditions for weathering tests.

Conditions	
Xenon arc lamp Borosilicate glass filters irradiance	6000 Watts 0.35 Watts per square meter at 340 nm
Procedure	
Exposure - arc lamp on	18 hours Black panel temp 63 ± 2°C Relative humidity 50 ± 2 percent (Water is sprayed onto the specimen for 18 minutes every 2 hours)
arc lamp off	6 hours Temperature 25 ± 2°C Relative humidity 90 to 95 percent
Total exposure	1200 hours

4.7.6.9 Fluid immersion (see 3.7.9).

4.7.6.9.1 Cable jacket material. Specimens of flat extruded cable outer jacket material shall be tested in accordance with EIA/TIA-455-12. The size of each specimen shall be in accordance with 4.7.5.15. After each 24 hour immersion, the specimen shall be removed, blotted to remove excess fluid, and suspended in the air at room temperature for not less than 3 hours, 30 minutes and not more than 4 hours, 30 minutes. The tensile strength and elongation of each specimen shall then be determined in accordance with 4.7.5.15.

4.7.6.9.2 Finished cable. One specimen of finished cable shall be tested in accordance with EIA/TIA-455-12. The two meter specimen may be cut into five pieces, and each piece immersed such that its two ends are exposed to the atmosphere. After each 24 hour immersion, the specimen shall be removed, blotted to remove excess fluid, and suspended in air at room temperature for not less than 3 hours, 30 minutes and not more than 4 hours, 30 minutes. After the test, the cable shall be tested for diameter change. The cable diameter shall be determined by measuring the cable circumference and dividing the resultant value by 3.14.

4.7.6.10 Wicking (see 3.7.10). Wicking characteristics shall be tested in accordance with EIA/TIA-455-39. The visual examination shall be made using ten-power magnification.

4.7.6.11 Jacket self-adhesion or blocking (see 3.7.11). Blocking characteristics shall be tested in accordance with EIA/TIA-455-84 to qualify the cable jacket self-adhesion property. The cable specimen shall be conditioned at the high storage temperature plus or minus 2°C for a period of 48 hours prior to testing for blocking. After the test exposure, the specimen shall be visually examined using ten-power magnification.

4.7.6.12 Flammability (see 3.7.12). Cables shall be tested in accordance with 4.7.6.12.1, 4.7.6.12.2, and 4.7.6.12.3 as specified.

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4.7.6.12.1 Flammability (60° angle) (see 3.7.12.1). Cables shall be tested as follows. The specimen shall be placed at an angle of 60 degrees with the horizontal and marked 200 mm from the lower end. The specimen shall be held taut in this position throughout the test using cable clamps or other suitable holding fixtures. A tissue shall be suspended horizontally 250 mm below the test specimen with the tissue center directly below the test mark on the specimen. A bunsen burner shall be adjusted to produce a 75 mm flame with the inner cone approximately 25 mm in height. The temperature of the hottest portion of the flame, as measured with a pyrometer, shall be not less than 950°C. The bunsen burner shall be positioned beneath the specimen (the burner shall be perpendicular to the specimen) so that the hottest portion of the flame is applied to the specimen at the test mark and the flame applied for 30 seconds. The time required for the specimen to self-extinguish, the maximum distance of flame travel from the mark, and any ignition of the tissue paper shall be recorded.

4.7.6.12.2. Flame extinguishing (see 3.7.12.2). Cables shall be subjected to the UL Flame Exposure test of UL-1685. Specimen length and quantity shall be as specified in UL-1685. The information specified in a through c shall be reported:

a. Flame temperature.

b. Period of time between burner shut off and cessation of flame on the specimen.

c. Overall distance of specimen jacket damage above the burner.

4.7.6.12.3 Smoke generation and flame propagation (see 3.7.12.3). Cable shall be tested for smoke generation and flame propagation in accordance with the Steniner Tunnel Test as described in UL-910.

4.7.6.13 Shock (see 3.7.13). A 30-meter minimum length of cable shall be used for this test. The specimen shall be subjected to grade A, type A, class I shocks as specified by MIL-S-901. Not less than 1 meter of the test specimen shall be mounted to simulate shipboard installation in a cable tray (refer to MIL-STD-2003, section 4 for guidance). After completion of the test, the cable shall be visually examined in accordance with 4.7.2. Signal discontinuity shall be monitored during the test, in accordance with TIA/EIA-455-32, with equipment having a time resolution sufficient to resolve discontinuities of duration not less than 50 μs.

4.7.6.14 Water absorption (see 3.7.14). Water absorption shall be determined using the gravimetric method of ASTM-D-470 with a water temperature of 71 $\pm$ 1°C for a continuous 3 day period. The exposed surface area of the finished cable or cable jacket specimen shall be not less than 5 square cm and not greater than 10 square cm.

4.7.6.15 Paint susceptibility (see 3.7.15). A two meter sample of finished cable shall be wrapped around a mandrel having an outer diameter equal to the minimum bend diameter of the cable. This cable mandrel assembly shall be painted using paint in accordance with TT-P-645 and allowed to dry for 120 hours. After drying the cable sample shall be removed from the mandrel and visually examined.

4.7.6.16 Electromagnetic effects (see 3.7.16). Cables shall be tested for conformance to 3.7.16 as specified herein. The cable shall be tested to determine the propagation characteristics (toward attenuation or conduction) of the cable through the waveguide. The enclosure shielding effectiveness for the enclosure to be used in the test shall be verified in accordance with the procedures specified in IEEE-STD-299 at the discrete frequencies specified in Table XVII with the waveguide in position. The dynamic range of the test setup shall be measured as specified in IEEE-STD-299 at each discrete frequency specified in Table XVII. The measured level of radio frequency (RF) propagation through the cable and waveguide installed in the shielded enclosure shall be determined in accordance with the procedures specified in IEEE-STD-299 at each discrete frequency specified in Table XVII.

TABLE XVII. Electromagnetic effects test frequencies.

Field propagation	Test frequency		
	1	2	3
Planewave	100 MHz	400 MHz	1 GHz
Microwave	10 GHz	---	---

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4.7.6.16.1 Cable positioning and setup. The cable specimen used shall be a minimum of 6 meters long. A minimum of 0.3 meters of cable shall be protruding from the waveguide on the receiver side of the enclosure (inside the shielded enclosure). The remaining cable length shall be extended normal to the enclosure wall on the transmitter side of the enclosure. The cable shall be suspended, dielectrically, at least 1 meter above the floor of the enclosure on both sides of the enclosure wall.

4.7.6.16.2 Waveguide type and configuration. The waveguide type used for this test shall be a waveguide at cut-off (WACO). The waveguide shall be installed in the setup panel or shielded enclosure wall and shall have an inside diameter to total length ratio of 1 to 72. For test purposes, a straight waveguide shall be used to simulate a worst case condition. The waveguide shall protrude inside the shielded enclosure from the wall.

4.7.6.16.3 Antenna orientation and setup. Antenna types and placement distances shall conform to Tables XVIII and XIX, respectively. Matched antennas shall be used for each measurement. The specified placement distances are for both antenna to cable and for antenna to setup panel or wall of the shielded enclosure. The transmitting antenna shall be oriented in both the parallel and perpendicular directions for each frequency listed in Table XVII. Energy from the transmitting antenna shall be maximized by positioning the antenna parallel along the cable length and perpendicular along the cable circumference. The sensitivity of the receiving antenna shall be maximized by measuring both end view and side (length) view with a constant rotation between the horizontal and vertical antenna positions. For planewave measurement in the perpendicular direction, the transmit antenna shall be 1 meter above the cable.

TABLE XVIII. Antenna types.

Field propagation	Antenna type
Planewave	Log periodic or dipole
Microwave	Horn

TABLE XIX. Antenna placement distances.

Location	Placement 1/
Receiving antenna to cable	5 cm
Transmitting antenna to cable	1 m
Transmitting antenna to shielded enclosure	2 m

1/ Extended dipole antenna distance measurements shall be made from the center of the antenna elements.

4.8. Chemical tests.

4.8.1 Acid gas generation (see 3.8.1). The total emission of any soluble acids (pH less than 3) shall be determined as follows. The required apparatus is shown on figure 4. A weighted sample of the finished cable, approximately 2.5 cm long, shall be placed in a silica boat which is put into the center of a silica tube, 26 to 60 cm long with 3.7 to 15.3 cm of internal diameter. The silica tube shall be placed in the tube furnace. An air supply, derived from a blower or compressed air cylinder, at the rate of 1 liter per minute, plus or minus 5 percent, shall be passed through the silica tube and then through four absorber flasks each containing 150 mil of deionized water. The furnace heating shall be commenced and the temperature of the tube and sample shall be raised to 800°C plus or minus 10°C over a period of approximately 40 minutes and then held at temperature for an additional 40 minutes. During the heating period, acid gases produced will be carried into the absorber flasks by the air flow. On completion of heating cycle, the fluids in the absorber flasks shall be titrated against 0.1 normal sodium hydroxide solution using Congo Red as an indicator. The total titer indicates the total soluble acids; 1.0 ml of 0.1 normal sodium hydroxide solution is equivalent to 3.65 mg of acid expressed as acid equivalent relative to hydrochloric acid.

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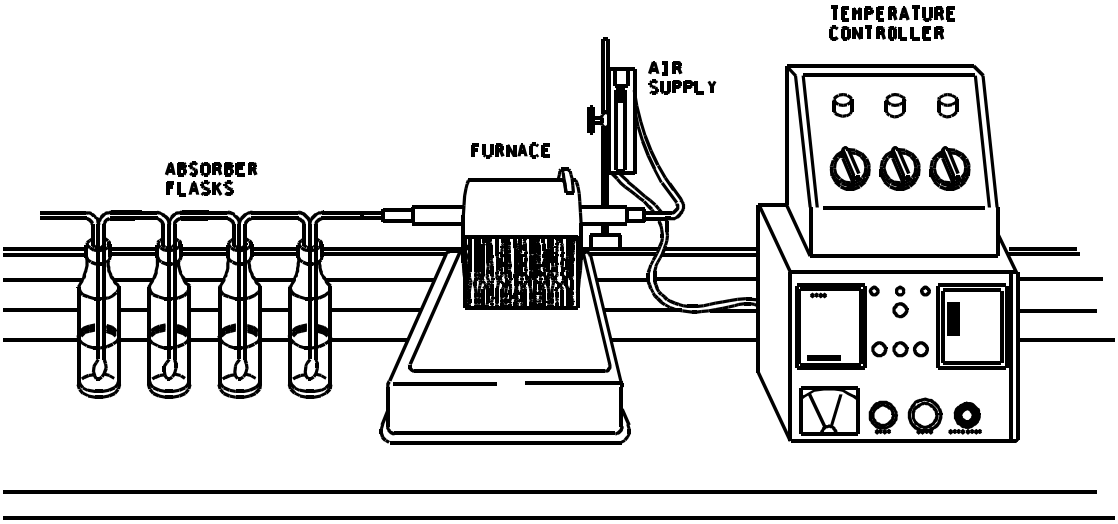


FIGURE 4. Acid gas generation test apparatus.

4.8.2 Halogen content (see 3.8.2). Cable specimens shall be tested to determine the fluorine, chlorine, bromine, and iodine content using any of the methods identified in 4.8.2.1 through 4.8.2.3.

4.8.2.1 X-ray fluorescence. An X-ray fluorescence instrument shall be used to determine the amount of all four halogens in an approximately 1 gram sample. The sample shall not be combusted prior to making the measurement. The X-ray fluorescence instrument used shall have sufficient measurement detection limits for each of the four halogens.

4.8.2.2 Ion chromatograph. Approximately 1 gram of finished cable shall be analyzed to determine the amount of all four halogens. Sample pretreatment shall be performed in accordance with ASTM-E-800. Halogen content shall be determined in accordance with ASTM-D-4327.

4.8.2.3 Ion selective electrode. Approximately 1 gram samples shall be used in these tests. Sample pretreatment shall be performed in accordance with ASTM-E-800. Fluoride content shall be determined in accordance with ASTM-D-1179, ASTM-D-3868, or ASTM-D-3761. Chloride content shall be determined in accordance with ASTM-D-512. Bromide content shall be determined in accordance with ASTM-D-1246. Iodide content shall be determined in accordance with method C of ASTM-D-3869.

4.8.3 Toxicity index (see 3.8.3). The toxicity index of the finished cable shall be tested in accordance with NES-713. An 8 cm length of cable shall be prepared in the following manner. Coat the ends of the exposed cable core material with a high temperature ceramic cement and allow 72 hours to cure. This capping procedure should eliminate dripping of the core material during testing. Support the cement caps with a rigid clamping device to hold the caps in place during testing. Position the cable sample above the flame source at a 45 degree angle and apply the flame so that it is centered on the cable sample. The combustion gases shall be chemically analyzed using calorimetric gas reaction tubes.

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4.8.4 Fungus resistance (see 3.8.4). Cables composed of materials not listed as fungus inert in guideline 4 of MIL-HDBK-454 shall be tested in accordance with TIA/EIA-455-56.

4.8.5 Materials test for space applications.

4.8.5.1 Thermal vacuum outgassing (see 3.3.6.1). Non-metallic materials shall be tested in accordance with ASTM-E-595.

4.8.5.2 Material flammability (see 3.3.6.2). Material samples shall be tested in accordance with NASA Handbook NHB 8060.1, test 1.

4.8.5.3 Material toxicity (offgassing) (see 3.3.6.3). Material samples shall be tested in accordance with NASA Handbook NHB 8060.1, test 7.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of material is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Points' packaging activity within the Military Department or Defense Agency, or within the military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature which may be helpful, but is not mandatory.)

6.1 Intended use. The fiber optic cables covered by this specification are intended for use in military applications where their performance characteristics are required. These cables are military unique due to the fact that these cables must operate satisfactorily in demanding system environments as listed below. For example, shipboard application conditions include numerous unique requirements for cables such as fluid, shock, stringent toxicity and flammability requirements.

a. Fixed plant. Used in systems in fixed locations including indoor, outdoor aerial, direct burial, duct, and undersea applications.

b. Tactical. Concerned with use in non-vehicular and mobile militarized systems.

c. Space. Which involves use in vehicles and/or systems deployed in outer space.

d. Avionics. Involving use in aircraft or missile systems.

e. Shipboard. Involving use in systems deployed in a mobile marine environment (on board or in tow).

f. Ground vehicle. Involving use in land vehicular systems.

g. Other specialized military applications.

6.1.1 Temperature rating. Temperature ratings as specified in specification sheets pertaining to this specification represent the maximum permissible operating temperature range of the cable.

6.1.2 Materials compatibility. The jacketing systems of the fiber optic cables covered by this specification may be degraded by certain fluids or compounds. If such degradation occurs, the fluids or compounds and the conditions necessary for failure shall be added to the specification sheet as a precautionary note.

6.1.2.1 Compatibility note. The insulation systems of polyvinylidene fluoride jacketed cables of this specification may be degraded by contact with hydraulic fluids of the phosphated ester type at high temperature. Cables of this specification with polyvinylidene fluoride jackets are not recommended for applications where they will be in contact with hydraulic fluids of phosphate ester types at temperatures above 50°C.

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6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.

b. Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1 and MIL-PRF-85045F).

c. Title, number, and date of the applicable specification sheet.

d. Complete cable part number.

e. Jacket and fiber color required (see 3.4.2 and 3.4.3).

f. Special marking requirements.

g. Level of packaging and packing required (see 5.1).

h. Minimum acceptable continuous cable length (see 3.1).

i. Total quantity of cable required.

j. Meter marking requirements (see 3.1 and 3.9).

k. Equivalent test methods, if other than as specified (see 4.7.1).

l. Length of cable per reel.

m. Data required.

n. White outer jacket color (if not black).

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in the applicable Qualified Products List QPL-85045, whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government, tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Commander, Naval Sea Systems Command, (SEA 03R42), 2531 Jefferson Davis Highway, Arlington, VA 22242-5160; however, information pertaining to qualification of products may be obtained from the agent for the qualifying activity: Defense Supply Center Columbus, (ATTN: DSCC-VQ), 3990 East Broad Street, Columbus, OH 43216-5000 (see 3.2 and 4.5).

6.3.1 Provisions governing qualification SD-6. Copies of "Provisions Governing Qualification SD-6" may be obtained upon application to Standardization Documents Order Desk, Bldg 4D, 700 Robbins Ave Philadelphia, PA 19111-5094.

6.3.2 Verification program. A verification program must be established and maintained in accordance with MIL-STD-790 or equivalent standard. Evidence of such compliance will be verified by the qualifying activity of this specification as a prerequisite for qualification and continued qualification. The verification system procedures, planning, and all other documentation and data that comprise the verification system must be available to the Government for review. The Government may perform any necessary inspections, verifications, and evaluations to ascertain conformance to the requirements and the adequacy of the implementing procedures (see 4.1).

6.3.3 Test equipment and inspection facilities. Provision for test and measuring equipment and inspection facilities of sufficient accuracy, quality, and quantity to permit performance of the required inspections must be the responsibility of the contractor. The establishment and maintenance of a calibration system to control the accuracy of the measuring and test equipment must be in accordance with NCSL Z540-1 or equivalent standard (see 4.2).

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6.3.4 Alternative forms of conformance inspection and equivalent test methods. Requests for alternate forms of conformance inspection (see 4.6) must be submitted to the qualifying activity and to the preparing activity. Alternate forms of conformance inspection may be used upon written approval by the qualifying activity and by the preparing activity. The use of equivalent test methods is allowed (see 4.7.1). The manufacturer must have conducted both test methods and have submitted complete test data to the preparing activity and to the qualifying activity verifying the equivalency of each alternate test method proposed.

6.3.5 Conformity to qualified sample. It is understood that cable supplied under contract must be identical in every respect to the qualification sample tested and found satisfactory, except for changes previously approved by the Government. Any unapproved changes from the qualification sample will constitute cause for rejection.

6.3.6 Forwarding of qualification samples. Samples and the manufacturer's certified test reports must be forwarded to the testing laboratory designated in the letter of authorization from the activity responsible for qualification (see 6.3). Samples will be plainly identified by securely attached, durable tags marked with the following information:

a. Sample for qualification test.

b. "MIL-PRF-85045F".

c. Specification sheet part number.

d. Manufacturer's name and CAGE code number.

e. Manufacturer's part number.

f. Comprehensive description and prime manufacturer's name and formulation number of the base materials from which the product is made. (This information will not be divulged by the Government.)

g. Place and date of manufacture of sample.

h. Submitted by (name) (date) for qualification tests in accordance with the requirements of MIL-PRF-85045 under authorization (reference authorizing letter).

6.3.7 Optical transmittance instrumentation stability. Optical transmittance instrumentation should be subjected to the following stability tests before qualification testing is performed. The first test should consist of measuring the transmitted power through each channel once every minute for a four hour period. The second test should consist of measuring the transmitted power through each channel once every 30 minutes for a 96 hour period. The data for each channel should be analyzed to determine average transmittance, minimum and maximum transmittance, the standard deviation of the transmittance, and the minimum and maximum percent deviation of the transmittance.

6.3.8 Government witnessing of qualification tests. The qualification activity may require that a government witness be present during the following tests: tensile loading and elongation, cyclic flexing, low temperature flexibility (cold bend), impact, cable twist-bending, hosing (low pressure and hydrostatic), cable jacket material tensile strength and elongation, thermal shock, fluid immersion, water absorption, flammability, cable abrasion resistance, and cross-link verification.

6.3.9 Electromagnetic effects test documentation. Electromagnetic effects test documentation should include a detailed test plan and test report. The test plan should detail the specific test setups and procedures, the test facility, and the test profiles that will be used. The test report should detail the exact procedures followed, the equipment used for each test phase, calibration dates of all test equipment, test results in graphical and tabular format, photographs/sketches of the test setups, the test data sheets, and an indication of pass/fail. The test data sheets should include test sample identification, ambient temperature and humidity values, dates and times of the test initiation and completion, names and initials of the test personnel, and data tables. The data tables should include the frequency at which each measurement was conducted, the field propagation/orientation, enclosure shielding effectiveness, reference level of the transmitter, dynamic range of the measurement, measured level for the test, the propagated RF attenuation, the required RF attenuation, and an indication of pass/fail. The reference level should include the source output, receiver sensitivity, and antenna gain.

6.3.10 Rejected production units. If a production unit is rejected, the supplier may screen out the defective units of product (if possible), and resubmit for reinspection.

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6.3.11 Noncompliance. If a specimen fails to pass group C inspection, the manufacturer shouldl notify the qualifying activity and the cognizant inspection activity of such failure and take corrective action on the materials or processes, or both, as warranted. Acceptance and shipment of the product should be discontinued until corrective action, acceptable to the qualifying activity, has been taken. After the corrective action has been taken, group C inspection shouldl be repeated on additional sample units (all tests and examinations, or the test which the original sample failed, at the option of the qualifying activity). Group A and group B inspections may be reinstituted; however, final acceptance and shipment shouldl be withheld until the group C inspection has shown that the corrective action was successful. In the event of failure after reinspection, information concerning the failure should be furnished to the cognizant inspection activity and the qualifying activity.

6.4 Personnel safety. Care should be taken when handling the very fine (small diameter) optical fibers to prevent skin puncture or contact of fiber with the eye area. Also, direct viewing of the optical terminal face of a terminated cable, while it is propagating optical energy, is not recommended unless the radiation is in the visible portion of the optical spectrum, of sufficiently low power, and needed to perform test examinations not obtainable by other methods.

6.5 Definitions. The following definitions of terms in this document are generally accepted by the optical fiber cable manufacturing industries. EIA-440 may be used as an additional reference for definitions of terms related to fiber optics.

6.5.1 Binder. A binder is a string or tape which ties together a number of fibers, buffered fibers, or OFCC's.

6.5.2 Buffered fiber. A buffered fiber is a coated optical fiber augmented with an additional coating or buffer jacket to protect the fiber and render it more visible and manageable.

6.5.3 Cable bundle. A cable bundle is a number of fibers, buffered fibers, ribbons, or OFCC's, grouped together in the cable core within a common protective layer.

6.5.4 Cable bundle jacket. A cable bundle jacket is the material which forms a protective layer around a bundle of fibers, buffered fibers, ribbons or OFCC's.

6.5.5 Cable core. Cable core is the part of the cable interior to the outermost jacket.

6.5.6 Cable core component. Cable core component is a part of the cable core, such as a buffered fiber, OFCC, cable bundle, ribbon and perhaps other parts.

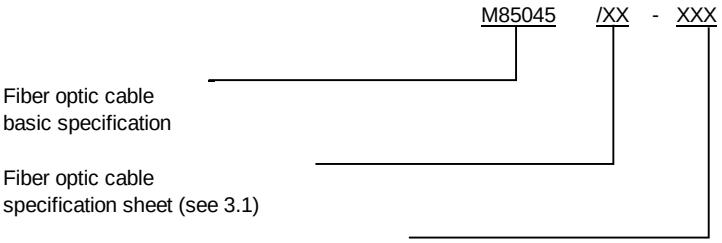
6.5.7 Long term minimum bend diameter. Long term minimum bend diameter is the minimum diameter at which a cable may be bent for extended periods of time with no degradation in optical performance.

6.5.8 OFCC. An OFCC is a buffered fiber augmented with a concentric layer of strength members and an overall jacket.

6.5.9 Ribbon. A ribbon is optical fibers arrayed side by side and maintained in this lateral position by various means.

6.5.10 Short term minimum bend diameter. Short term minimum bend diameter is the minimum diameter at which the cable may be bent for short periods of time (such as during cable installation) with no permanent degradation in optical performance.

6.6 Part or Identifying Number (PIN). The PIN shall be constructed in accordance with the following:



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6.7 Subject term (key word listing).

Aperture, numerical
Armor sheathing
Attenuation
Bandwidth
Binder
Bundle
Core
Crosstalk
Diameter, cable
Dielectric construction
Flammability
Jacket
OFCC
Ribbon

6.8 Changes from previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Army - CR
Navy - SH
Air Force - 11

Review activities:
Army - AR, AV, MI,
Navy - EC, OS
Air Force - 02, 13, 19, 80, 99
NASA - NA

Preparing activity:
Navy - SH

Agent:
DLA - CC

(Project 6015-0034)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

2. The submitter of this form must complete blocks 4, 5, 6, and 7, and send to preparing activity.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	**1. DOCUMENT NUMBER** MIL-PRF-85045F	**2. DOCUMENT DATE (YYMMDD)**

3. DOCUMENT TITLE
CABLES, FIBER OPTICS, (METRIC) GENERAL SPECIFICATION FOR

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) DSN (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME Claude Cassidy	b. TELEPHONE (Include Area Code) (1) Commercial (703) 602-7468 X-279 (2) DSN 332-7468 X-279
c. ADDRESS (Include Zip Code) Naval Sea Systems Command 03R42 2531 Jefferson Davis Highway Arlington, VA 22242-5160	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** Defense Standardization Program Office (DLSC-LM) 8725 John J. Kingman Road, Suite 2533 Fort Belvoir, Virginia 22060-6221 Telephone (703) 767-6888 DSN 427-6888

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